Opening the Black Box of Logical Remainders: A Transparency Reporting Framework for fsQCA Intermediate Solutions

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Abstract

Fuzzy-set qualitative comparative analysis (fsQCA) intermediate solutions rely on logical remainders and directional expectations, yet conventional outputs usually report only final formulas and fit indices. They rarely show how remainders are incorporated, excluded, or affected by prime implicant (PI) cover multiplicity. The framework developed here formalizes term-pair extension sets $\boldsymbol{E(c,s)}$, cover-pair admitted-remainder sets $\boldsymbol{E_{ij}}$, and excluded-remainder categories $\boldsymbol{G_{ij}^{A}}$ and $\boldsymbol{G_{ij}^{B}}$. It also summarizes direction conflicts, path stability, and minterm-status stability across cover-pair records. A digital transformation case shows how the report complements standard fsQCA output by making cover-pair sensitivity and remainder-status variation visible. The framework leaves minimization procedures unchanged and treats the proposed categories as disclosure tools rather than solution-selection criteria. Its value lies in making the counterfactual support structure behind intermediate solutions more transparent and auditable.

Keywords: fsQCA, qualitative comparative analysis, intermediate solution, logical remainders, PI-cover multiplicity, transparency reporting framework

1 Introduction

Qualitative comparative analysis (QCA) was introduced as a comparative method for analyzing configurational causation beyond a strict division between qualitative and quantitative strategies (Ragin 1987). Fuzzy-set qualitative comparative analysis (fsQCA) extends this logic by assigning cases calibrated degrees of set membership and organizing them into truth-table rows (Ragin 2000). Because the number of logically possible rows increases rapidly with the number of conditions, empirical data rarely cover the full Boolean space. The unobserved rows are logical remainders. Their treatment is central to fsQCA because complex, parsimonious, and intermediate solutions differ in part by how they use, restrict, or avoid such rows. Intermediate solutions are especially important in this respect: they rely on directional expectations to decide which counterfactual rows can be used for simplification (Ragin 2008; Schneider and Wagemann 2012). Logical remainders are therefore not a peripheral technical detail, but part of how fsQCA connects limited empirical diversity with theoretically informed counterfactual reasoning.

Several strands of QCA scholarship have clarified adjacent parts of this process. Software-oriented work has made calibration, truth-table construction, Boolean minimization, and solution derivation operationally available for applied research (Thiem and Dusa 2013; Dusa 2019). Enhanced Standard Analysis and related procedures address the plausibility of simplifying assumptions before or during solution construction (Oana and Schneider 2018). Robustness research examines how QCA results respond to calibration choices, threshold settings, and specification changes (Skaaning 2011; Oana and Schneider 2024). Work on minimization and model ambiguity further shows that alternative minimal solutions or causal models can arise from the same configurational data (Dusa 2018; Baumgartner and Thiem 2017). Taken together, these studies have improved how QCA solutions are generated, assessed, tested, and qualified.

The remaining issue is not located at these earlier stages of solution construction, but at the point where an intermediate solution is reported. Once the truth table, directional expectations, and relevant complex-cover and parsimonious-cover alternatives have been determined, conventional output usually condenses the result into a simplified expression and fit indices. This compression makes the intermediate procedure only partly visible from the final output itself. Prior formal work has partly addressed the procedural side of this opacity by reconstructing fsQCA as a sequence of formal steps, showing that mathematical reconstruction can make important features of the procedure more explicit (Mendel and Korjani 2012, 2013). However, this work mainly clarifies how fsQCA solutions are produced; it does not turn the logical remainders behind intermediate solutions into systematic reporting objects. As a result, conventional output still does not normally show how logical remainders are incorporated into, or excluded from, an intermediate solution. This matters especially when cover-pair alternatives rely on different counterfactual supports while producing identical or closely related final expressions. In such cases, the final formula can conceal variation in the remainder-use structure behind the solution.

The logical-remainder transparency report developed here targets this fixed-output stage. The framework does not change standard minimization procedures or replace existing QCA software. Instead, it turns the downstream admission and exclusion of logical remainders into reportable objects. Specifically, it formalizes the term-pair extension set $E(c, s)$, the cover-pair admitted-remainder set E_{ij}, and the within-boundary and outside-boundary excluded-remainder categories G_{ij}^A and G_{ij}^B. It also summarizes direction conflicts, intermediate-expression/path stability, and logical-remainder/minterm-status stability across complex-cover and parsimonious-cover alternatives.

The contribution is therefore not a new fsQCA algorithm, but a reporting layer that extends prior efforts to formalize fsQCA procedures from procedural description to reporting transparency. Its purpose is to make the counterfactual support structure behind intermediate solutions explicit enough to be inspected, reproduced, and appropriately interpreted in applied reporting. On this basis, the study is guided by three research questions:

RQ1. How can the admission and exclusion of logical remainders in fsQCA intermediate solutions be transparently represented?

RQ2. How are complex-cover and parsimonious-cover alternatives associated with variation in logical-remainder status and minterm stability, including admitted remainders, G_{ij}^A, and G_{ij}^B?

RQ3. How can these differences be organized into a reproducible logical-remainder transparency report?

2 Background and Related Work

2.1 Logical Remainders and Intermediate Solutions in fsQCA

Logical remainders become important in fsQCA because they mark the point at which empirical limited diversity and theoretical counterfactual reasoning meet. In fsQCA, a truth table assigns observed cases to combinations of calibrated conditions and classifies rows according to outcome values and threshold rules (Dusa 2019). Some observed rows are treated as supporting the outcome, whereas other observed rows do not provide sufficient support for it. By contrast, logical remainders are rows for which the logically possible combination of conditions lacks sufficient empirical instances. They are therefore not simply missing observations or empty cells in a table. Rather, they express the fact that empirical cases occupy only part of the logically possible Boolean space.

The treatment of these unobserved rows is one of the main differences among complex, parsimonious, and intermediate solutions (Ragin 2008; Schneider and Wagemann 2012). The complex solution avoids using logical remainders and therefore stays closest to the observed positive rows. The parsimonious solution allows remainders to enter minimization more freely as simplifying assumptions, which can produce shorter and more general expressions. The intermediate solution occupies a more constrained position. It uses counterfactual rows only when their use is consistent with the researcher's directional expectations (Ragin and Sonnett 2005). Directional expectations are therefore not an optional annotation attached to the solution; they are the means through which substantive knowledge shapes which unobserved rows may support simplification.

This point has direct implications for how intermediate solutions should be reported. An intermediate formula is a compact term-level representation of a Boolean region. Each solution term, or path, covers a set of truth-table rows; at the minterm level, these rows may include observed positive configurations, logical remainders admitted as counterfactual support, or both. In this sense, logical remainders are not external to the solution. They are part of the minterm-level support structure through which a simplified term becomes possible.

This distinction matters because Boolean expressions can sometimes be written in equivalent or closely related forms, whereas the truth-table rows covered by those expressions provide a more basic way to inspect what the solution rests on. Fit indices are also necessary, but they summarize consistency and coverage at the level of the reported solution; they do not identify how each term is anchored in observed and counterfactual rows. A final intermediate formula may therefore show the simplified result without showing the remainder structure that helped produce it.

For this reason, the final intermediate formula and fit indices are incomplete as reporting objects. They tell readers which simplified expression has been obtained and how well it fits the observed cases, but they do not disclose which logical remainders were incorporated into the simplification, which potentially relevant remainders remained unused, or how observed and counterfactual minterms jointly support the reported terms. The reporting gap therefore begins with the difference between a solution formula as a compact term-level result and the minterm-level support structure that gives that result its counterfactual basis.

2.2 Prime Implicants, PI Charts, and PI-Cover Multiplicity

The incompleteness of formula-level reporting becomes more consequential once the minimization process is considered from the perspective of prime implicants and PI-cover selection. Boolean minimization in QCA builds on the logic of simplifying truth functions (Quine 1952; McCluskey 1956). In QCA applications, truth-table rows provide the input for minimization, while prime implicants serve as candidate simplified terms from which solution expressions are derived (Dusa 2018). Software implementations have made these steps operationally available by supporting calibration, truth-table construction, and the derivation of complex, parsimonious, and intermediate solutions (Thiem and Dusa 2013). Yet the availability of a final solution expression does not remove the need to inspect how the expression was assembled from alternative implicant choices.

A PI chart makes this selection problem explicit. Its rows represent prime implicants, and its columns represent the positive truth-table rows that must be covered. Some prime implicants are essential because they cover a target row that no other candidate covers. Others are substitutable: more than one implicant, or more than one combination of implicants, can complete the same covering task. This substitutability is one source of model ambiguity in configurational comparative research (Baumgartner and Thiem 2017). The important point for the present study is that PI-cover multiplicity can affect the regions of the Boolean space opened by a solution, even when alternative covers solve the same PI-chart covering problem.

This implication is not limited to parsimonious minimization. PI-cover alternatives may also arise in complex minimization, although they are often more visible in parsimonious minimization because logical remainders are available as don't-care rows. For this reason, the reporting objects below are defined over complex-cover and parsimonious-cover pairs rather than over a single fixed complex cover. This specification locates the level at which admitted logical remainders are induced.

This has direct consequences for the reporting of logical remainders. A cover-pair alternative may preserve the same required coverage of observed positive rows while changing the unobserved rows that become relevant to the resulting intermediate expression. In other words, the ambiguity is not only formula-level ambiguity. It can also be a support-structure ambiguity: different cover-pair records may rely on different combinations of observed positive minterms and admitted logical remainders, while leaving other directionally plausible remainders unused. This means that two alternatives may produce identical or closely related final expressions, but still differ in how paths are supported, which remainders are incorporated, and which minterms change status across covers.

For this reason, PI-cover multiplicity should not be reported only as alternative covers or alternative formulas. Its consequences for the structure behind the formula also need to be made visible. At a minimum, this requires attention to whether solution paths remain stable across covers, whether the same logical remainders are admitted or excluded, and whether minterms retain the same observed, counterfactual, or unadmitted status. From this perspective, cover-pair alternatives deepen the reporting problem identified above: they show that the final expression and fit indices may summarize the result while still concealing variation in the path-level and minterm-level support structure that makes the intermediate solution possible.

Because remainder admission is evaluated after complex-cover and parsimonious-cover alternatives have been generated, the row-level support of an intermediate solution can become conditional on the selected cover pair and the effective term-pair structure. The reporting framework developed below makes this ordering consequence inspectable without altering the standard procedure.

2.3 Robustness, Sensitivity, and Reporting Transparency in QCA

Once intermediate solutions are viewed as compact representations of an underlying minterm-level support structure, and once PI-cover multiplicity is understood as a possible source of variation in that structure, the reporting problem can be positioned more precisely in relation to existing QCA scholarship. Table 1 summarizes this positioning by contrasting the analytical focus of standard QCA outputs, enhanced standard analysis, robustness research, and PI-chart/model-ambiguity research with the support-structure reporting problem addressed here.

Standard QCA reporting provides essential solution-level outputs. Complex, parsimonious, and intermediate formulas identify the expressions obtained from minimization (Ragin 2008). Consistency and coverage indicators evaluate the empirical strength of the reported set relations (Ragin 2006). QCA with R also treats truth tables, logical minimization, and pseudo-counterfactual analysis as core elements of the applied workflow (Dusa 2019). These outputs and procedures are indispensable for applied reporting because they establish the conventional solution, its fit to the observed cases, and the workflow through which the result is generated. At the same time, they primarily present the solution at the level of formulas, indices, and procedural outputs. They do not usually disclose how the reported intermediate expression is anchored at the row level in observed positive minterms, admitted logical remainders, and remainders that remain unused.

A related body of work addresses the credibility of simplifying assumptions and the sensitivity of QCA results. Enhanced standard analysis and counterfactual assessment examine the plausibility of simplifying assumptions before or during solution construction (Oana and Schneider 2018). Robustness checks examine how calibration, frequency thresholds, and consistency thresholds affect QCA results (Skaaning 2011). Parameter and specification sensitivity are central concerns in fsQCA robustness research (Krogslund et al. 2015). Combinatorial diagnostics further examine sensitivity in QCA reference solutions (Thiem et al. 2016). Recent robustness protocols organize sensitivity ranges, fit-oriented robustness, and case-oriented robustness for applied QCA (Oana and Schneider 2024). These studies strengthen QCA practice by examining the assumptions, thresholds, and specifications under which solutions are obtained. The reporting perspective developed here is complementary to this tradition. Rather than varying upstream calibration or threshold choices, it begins from a specified analytical configuration and asks how the counterfactual support structure compressed into the final intermediate expression can be made explicit.

Software, minimization, and model-ambiguity research clarify another adjacent dimension of the problem. QCA software supports calibration, truth-table construction, and solution derivation (Thiem and Dusa 2013). Exact Boolean minimization work improves the efficient derivation of complete prime-implicant results (Dusa 2018). Model-ambiguity research shows why alternative minimal unions or causal models should be made visible when they arise (Baumgartner and Thiem 2017). These contributions are important because they make the generation and representation of QCA solutions more explicit. However, cover-pair alternatives also raise a further reporting issue concerning their consequences for the solution's support structure. Alternative covers may preserve the required coverage of positive rows while changing which logical remainders are incorporated, which remain excluded, and which minterms change status. The relevant issue is therefore not only the existence of alternative covers, but also how their differences affect the path-level and minterm-level structure behind the intermediate solution.

This positioning provides the basis for the formal development in the next section. Section 3 treats conventional fsQCA outputs as fixed inputs and translates the support-structure problem into reportable objects. It defines how admitted and excluded logical remainders can be represented, how direction conflicts can be identified, and how path stability and minterm-status variation can be traced across cover-pair records. The purpose is to move from a general claim about reporting transparency to a reproducible framework for inspecting the counterfactual structure behind already generated intermediate solutions.

Table 1 Positioning of the proposed transparency report relative to existing QCA outputs and sensitivity research

Area	Existing focus	Less visible issue	Added value of this paper
Standard QCA solution output (Ragin 2008; Schneider and Wagemann 2012; Dusa 2019)	Solution formulas; consistency; coverage	Row-level remainder admission and exclusion	Reports admitted and excluded remainder statuses
Enhanced standard analysis and counterfactual assessment (Schneider and Wagemann 2012; Oana and Schneider 2018)	Plausibility of simplifying assumptions	Ex post remainder-status distribution	Reports downstream support structure
Calibration and threshold robustness (Skaaning 2011; Krogslund et al. 2015; Thiem et al. 2016; Oana and Schneider 2024)	Sensitivity to calibration and thresholds	Fixed-output internal support structure	Holds upstream inputs fixed
PI-chart and PI-cover reporting (Thiem and Dusa 2013; Dusa 2018, 2019; Baumgartner and Thiem 2017)	Prime implicants; cover alternatives	Remainder, path, and minterm status variation	Reports path stability and minterm-status changes

3 Logical-Remainder Transparency Framework

An intermediate solution can be read not only as a minimized formula, but also as a row-level structure supported by observed positive rows and admitted logical remainders. Under fixed truth-table inputs and directional expectations, the relevant reporting task is to make that support structure inspectable without changing the minimization result. This requires mapping solution terms back to the Boolean space, identifying the remainders admitted by the standard term-pair procedure under each complex-cover and parsimonious-cover pair, and then locating directionally reachable remainders that are not admitted. Because cover alternatives may open different Boolean regions and produce different admitted-remainder sets, the same remainder may require comparison across cover-pair records. Figure 1 summarizes this workflow before the formal components are defined in detail.

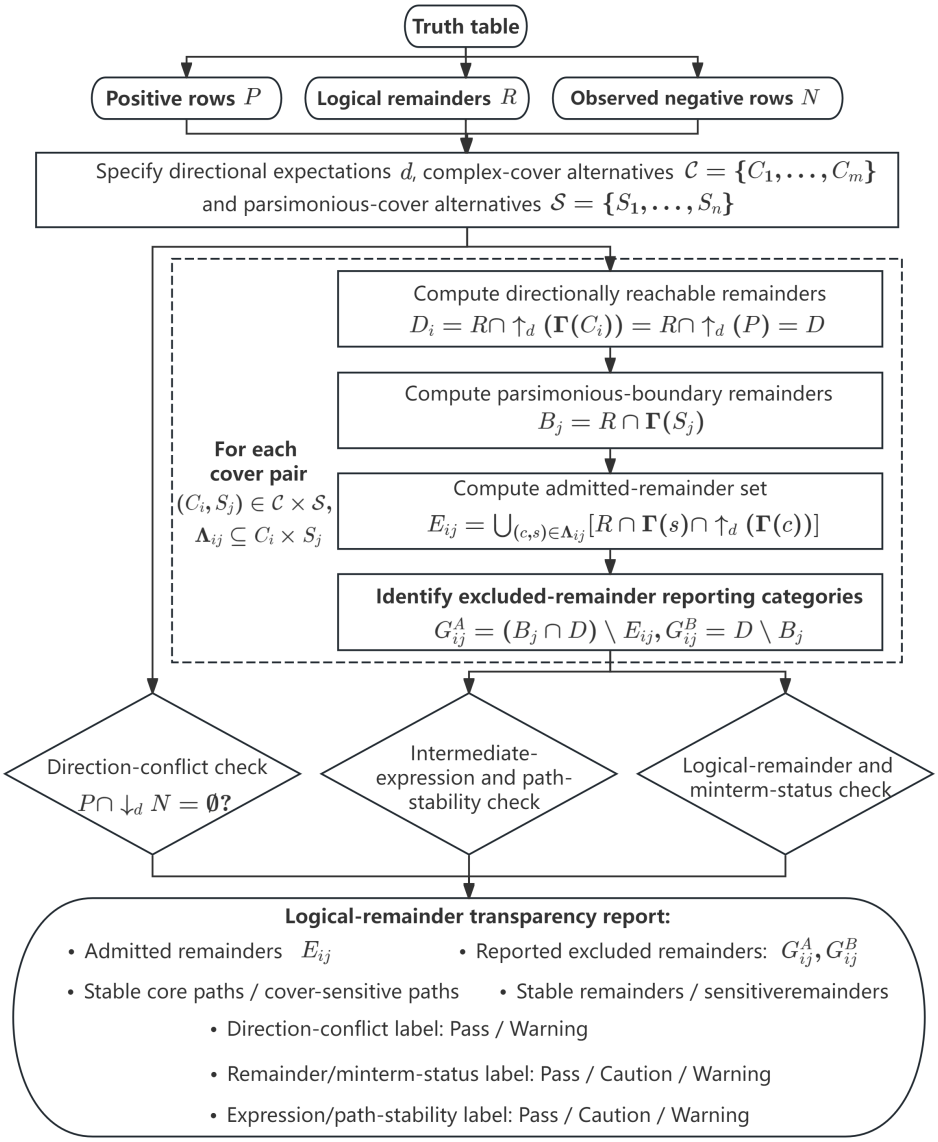

Fig. 1 Workflow of the logical-remainder transparency report. The workflow starts from fixed truth-table inputs, directional expectations, complex-cover alternatives, and parsimonious-cover alternatives, and turns admitted/excluded remainders, path stability, and minterm-status stability into reportable outputs.

3.1 Truth-Table Partition and Boolean Coverage

The first step is to restore the link between a minimized expression and the truth-table rows it covers. Let $\Omega = \{0,1\}^q$ denote the Boolean space for q causal conditions:

$$\Omega = \{0,1\}^q. \tag{1}$$

Each element of Ω is a minterm, or a complete truth-table row. The fixed truth table partitions this space into

$$P, \quad N, \quad R. \tag{2}$$

Here P denotes positive truth-table rows, N denotes observed negative rows, and R denotes logical remainders. These sets are pairwise disjoint and satisfy $P \cup N \cup R = \Omega$. The partition is treated as a fixed input to the transparency report. Calibration, threshold selection, and truth-table construction are not repeated; the task is to trace how the already generated solution relates back to observed positive rows, observed negative rows, and logical remainders.

For a Boolean term t, the coverage map is written as

$$\Gamma(t) = \{x \in \Omega : x \text{ is covered by term } t\}. \tag{3}$$

For a set of terms S, the covered region is

$$\Gamma(S) = \bigcup_{s \in S} \Gamma(s). \tag{4}$$

These formulas turn a compact solution expression back into a minterm-level object. Once term coverage is mapped to truth-table rows, the report can inspect not only whether a row is covered by a term, but also whether the covered row is a positive row, an observed negative row, or a logical remainder. This row-level coverage map is necessary before the analysis can ask which logical remainders have counterfactual relevance under the stated directional expectations.

The notation $\mathrm{QM}_\Omega(A, D, O)$ is used below as a compact shorthand for an exact Boolean minimization call over Ω. The first argument denotes rows minimized as positive, the second denotes optional don't-care rows when supplied, and the third denotes additional constrained rows when specified; in the constructions used here, this third argument is empty. In the standard intermediate-expression construction reported below, the admitted remainders have already been incorporated into the positive region, so the corresponding notation is $\mathrm{QM}_\Omega(P \cup E_{ij}, \varnothing, \varnothing)$.

3.2 Directional Expectations and Directional Reachability

Directional expectations provide the link between substantive theory and the treatment of logical remainders. They are represented by

$$d \in \{0, 1, \circ\}^q. \tag{5}$$

A value $d_j = 1$ means that movement from absence to presence of condition j can support the expected outcome. A value $d_j = 0$ means that movement from presence to absence can support the expected outcome. A value $d_j = \circ$ means that dimension j is not used to justify directional movement.

To make these directional restrictions explicit at the row level, the report writes them as a relation between minterms. This notation is introduced here for reporting purposes, while the procedural role of directional movement restrictions is consistent with the rule-based reconstruction of fsQCA intermediate-solution procedures by Mendel and Korjani (2012, 2013). For two minterms x and y, write $x \preceq_d y$ if y can be reached from x by moving only along the directions specified by d. The corresponding upward and downward directional closures are

$$\uparrow_d A = \{y \in \Omega : \exists x \in A,\ x \preceq_d y\}, \tag{6}$$

and

$$\downarrow_d A = \{x \in \Omega : \exists y \in A,\ x \preceq_d y\}. \tag{7}$$

The upward closure identifies rows that are directionally reachable from a source set and therefore can be treated as candidate counterfactual rows for the transparency report. The downward closure prepares the later direction-conflict check by identifying rows that can reach a target set under the same directional assumptions. Directional reachability therefore marks possible counterfactual relevance, but it does not show which remainders have actually been admitted by the standard intermediate-solution procedure. The directional-closure notation introduced here provides the row-level language for the term-pair characterization developed below.

3.3 Effective Term Pairs and Standard Admitted Remainders

Directional reachability identifies candidate counterfactual rows, but the reporting problem now becomes more specific: which of those remainders support simplification under the standard intermediate-solution procedure. Let the optimal complex-cover alternatives be

$$\mathcal{C} = \{C_1, \ldots, C_m\}, \tag{8}$$

and let the optimal parsimonious-cover alternatives be

$$\mathcal{S} = \{S_1, \ldots, S_n\}. \tag{9}$$

Each C_i is a set of complex terms, each S_j is a set of parsimonious terms, and m and n need not be equal. Because complex minimization is constructed without logical remainders, each complex-cover alternative covers the positive rows, $\Gamma(C_i) = P$. Complex-cover multiplicity does not alter the purpose of the framework, but it clarifies the level at which remainder admission is defined.

For a parsimonious cover S_j, the remainder portion of the parsimonious boundary is

$$B_j = R \cap \Gamma(S_j), \tag{10}$$

where $\Gamma(S_j) = \bigcup_{s \in S_j} \Gamma(s)$. In remainder-status reporting, B_j therefore denotes the logical remainders covered by the selected parsimonious cover, not the full Boolean span itself.

The standard intermediate-solution procedure operates locally through complex–parsimonious term pairs. For each cover pair (C_i, S_j), define the effective term-pair set as

$$\Lambda_{ij} = \Lambda(C_i, S_j) = \{(c, s) \in C_i \times S_j : \Gamma(c) \subseteq \Gamma(s)\}. \tag{11}$$

An effective pair (c, s) means that the parsimonious term s absorbs the complex term c. This pair has a specific reporting interpretation: the complex term supplies the observed positive source region, while the parsimonious term supplies the local boundary within which counterfactual simplification is considered.

Let $A(c, s)$ denote the local rule-based acceptable-remainder set generated by the standard intermediate-solution counterfactual procedure for an effective pair (c, s). Appendix A translates the seven literature-based procedural rules behind this local admission step into the notation used here. In set-theoretic form, those rules impose three restrictions on a candidate remainder: it must be a logical remainder, it must remain inside the local parsimonious boundary, and it must be directionally reachable from the complex source term. Hence, for each effective pair, define the term-pair extension set

$$E(c, s) = R \cap \Gamma(s) \cap \uparrow_d (\Gamma(c)). \tag{12}$$

$E(c, s)$ denotes the logical remainders through which c can be directionally extended toward s. The object is defined at the term-pair level because remainder admission depends not only on the selected parsimonious boundary but also on the complex term that anchors the counterfactual extension.

The term-pair characterization below builds on the rule-based account of Ragin's fsQCA intermediate-solution procedure reconstructed by Mendel and Korjani (2012). In row-level terms, the standard admission logic requires a candidate remainder to satisfy the three conditions represented in Equation (12): remainder status, local boundary membership, and directional reachability from the complex source term. Equation (12) is therefore a reconstruction of standard admission for reporting purposes, not a new intermediate-solution algorithm.

Proposition 1. Term-pair characterization of admitted remainders. Conditional on a fixed truth-table partition (P, N, R), a fixed directional-expectation vector d, and a fixed cover pair (C_i, S_j), the local rule-based acceptable-remainder set and the set-theoretic term-pair extension set coincide:

$$A(c, s) = E(c, s) = R \cap \Gamma(s) \cap \uparrow_d (\Gamma(c)) \quad \text{for every } (c, s) \in \Lambda_{ij}. \tag{13}$$

The procedural rules behind this characterization and the proof are provided in Appendix A.

The admitted-remainder set induced by the cover pair (C_i, S_j) is

$$E_{ij} = \bigcup_{(c,s) \in \Lambda_{ij}} E(c, s) = \bigcup_{(c,s) \in \Lambda_{ij}} \left[R \cap \Gamma(s) \cap \uparrow_d (\Gamma(c))\right]. \tag{14}$$

E_{ij} records the logical remainders admitted by the standard complex–parsimonious term-pair procedure under the cover pair (C_i, S_j). It keeps the standard admission result as the object being reported. Here, "admitted" refers to this standard term-pair admission step, not to a separate normative judgment that the remainder should be interpreted as substantively causal. The row-level support for the corresponding intermediate expression is $P \cup E_{ij}$, the expanded positive region after standard counterfactual admission. The expression is then obtained by exact Boolean minimization:

$$I_{ij} = \mathrm{QM}_\Omega(P \cup E_{ij}, \varnothing, \varnothing). \tag{15}$$

At this final step, E_{ij} is not supplied as optional don't-care rows; its members have already been accepted through the directional counterfactual procedure and incorporated into the positive input to be minimized. The empty second and third arguments indicate that no additional don't-care rows are introduced in this final minimization step.

3.4 Within-Boundary and Outside-Boundary Excluded Remainders

Excluded remainders require a further distinction because exclusion can arise in different structural positions. Although E_{ij} may vary with the complex-cover decomposition, the directionally reachable logical-remainder set is common across complex-cover alternatives because each C_i covers the same positive rows P. Accordingly, define

$$D = R \cap \uparrow_d P. \tag{16}$$

Equivalently, for any C_i, $R \cap \uparrow_d (\Gamma(C_i)) = R \cap \uparrow_d P = D$. Different cover pairs may change which elements of D are admitted, but they do not change the directionally reachable logical-remainder set itself.

For a cover pair (C_i, S_j), the two excluded-remainder reporting categories are

$$G^A_{ij} = (B_j \cap D) \setminus E_{ij}, \tag{17}$$

and

$$G^B_{ij} = D \setminus B_j. \tag{18}$$

G^A_{ij} identifies remainders that are directionally reachable and lie inside the parsimonious boundary, but are not admitted through the effective term-pair extensions. G^B_{ij} captures directionally reachable remainders that remain outside the selected parsimonious boundary. Both categories are ex post reporting statuses rather than admission rules, solution-selection criteria, or substantive validity judgments.

For each cover pair, $E_{ij} \subseteq B_j \cap D \subseteq D$. The first inclusion holds because every term-pair extension set is restricted to R, $\Gamma(s)$, and directional reachability from a complex source term; since $\Gamma(C_i) = P$, reachability from $\Gamma(c)$ implies reachability from P. The second inclusion follows from the definition of $B_j \cap D$ as the boundary-restricted part of the common directionally reachable universe. Thus E_{ij}, G^A_{ij}, and G^B_{ij}

partition D for a given cover pair. Figure 2 illustrates how admitted remainders and the two excluded-remainder categories are related.

Under fixed P and d, cover-pair selection can change the parsimonious boundary B_j and the admitted-remainder set E_{ij}. The same directionally reachable remainder can therefore be admitted under one cover pair, remain inside the boundary but unadmitted under another, or fall outside the selected boundary under a third. This admitted/G_{ij}^A/G_{ij}^B variation is the row-level status switching that the later remainder/minterm-status check is designed to summarize. The set-theoretic objects defined here thus provide the basis for the structural checks that follow.

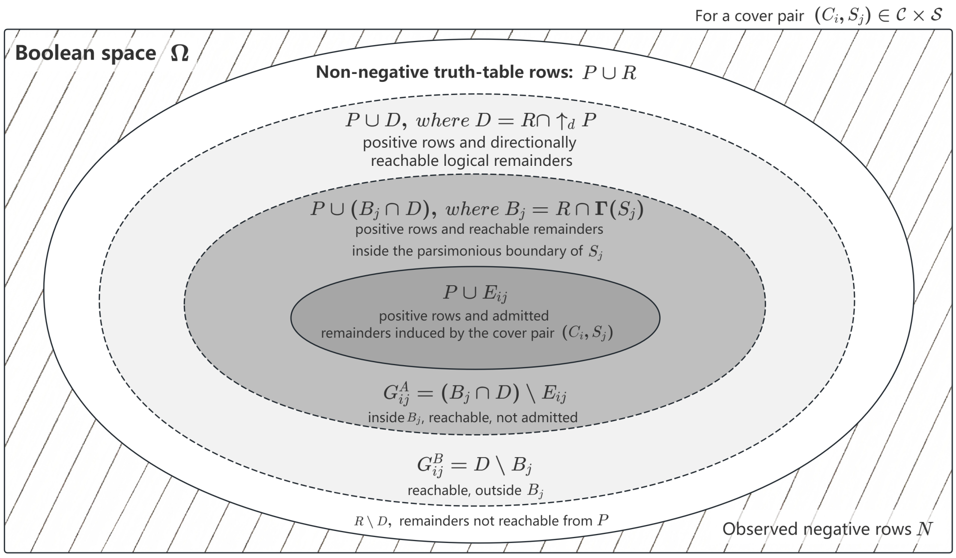

Fig. 2 Logical-remainder regions in the transparency report. For a cover pair (C_i, S_j), E_{ij} contains remainders admitted by the standard complex–parsimonious term-pair procedure. G_{ij}^A records directionally reachable remainders that fall inside the selected parsimonious boundary B_j but are not admitted. G_{ij}^B records directionally reachable remainders outside B_j. These regions are ex post reporting categories for the support structure behind the intermediate solution.

Together, P, N, R, directional reachability, $E(c, s)$, E_{ij}, B_j, D, G_{ij}^A, and G_{ij}^B define the row-level and cover-pair objects needed for the transparency report. Supplementary Material S1 establishes the formal properties that justify using these objects as bounded reporting categories: the directional relation $\preceq_d$ forms a partial order and supports monotone upward and downward closures; the direction-conflict check can be expressed equivalently through $P \cap \downarrow_d N = \emptyset$ or $\uparrow_d P \cap N = \emptyset$; E_{ij}, G_{ij}^A, G_{ij}^B, and non-directionally reachable remainders partition the logical-remainder space; and the admitted-remainder family supports infimum/supremum summaries under set inclusion. Appendix A gives the procedural-rule mapping and proof of the term-pair characterization. These row-level and cover-pair objects can then be organized into structural checks for direction conflict, expression/path stability, and remainder/minterm-status variation.

4 Three Structural Checks in the Transparency Report

The row-level and cover-pair objects defined above become interpretable when they are organized into reporting checks. The transparency report therefore uses three structural checks: a direction-conflict check, an intermediate-expression/path-stability check, and a logical-remainder/minterm-status check. Together, these checks address what final formulas and fit indices cannot disclose: whether directional expectations conflict with observed negative rows, whether cover alternatives generate expression or path sensitivity, and whether remainders or minterms switch among admitted, G_{ij}^{A}, and G_{ij}^{B} statuses. The label conditions are summarized in Table 2 after the checks are defined, and Figure 3 shows how these labels guide subsequent reporting actions.

4.1 Direction-Conflict Check

The direction-conflict check is

$$P \cap \downarrow_d N = \emptyset. \tag{19}$$

Using the directional-reachability relation defined above, this check asks whether any positive row can directionally reach an observed negative row. Because intermediate solutions rely on directional expectations to justify counterfactual simplification, compatibility between these expectations and observed negative rows becomes a necessary reporting concern (Ragin 2008; Schneider and Wagemann 2012). The check has only two possible reporting labels. If no conflict exists, the reporting label is Pass. If a conflict exists, the reporting label is Warning; there is no Caution category because the conflict directly concerns the compatibility between directional expectations and observed negative rows. A Warning label indicates that conflicting rows should be disclosed and that directional expectations, negative cases, or calibration choices should be reviewed.

4.2 Intermediate-Expression and Path-Stability Check

Let

$$K = \text{number of cover-pair records} \tag{20}$$

and

$$L = \text{number of unique intermediate expressions.} \tag{21}$$

Conditional on the fixed truth table, directional expectations, and minimization structure, cover-pair records can be associated with different intermediate expressions even when they cover the same positive truth-table rows. The need for this check follows from minimization and model-ambiguity research showing that alternative prime-implicant covers or minimal models can arise from the same configurational data (Dusa 2018; Baumgartner and Thiem 2017). If $K = 1$, the expression-stability reporting label is Pass. If $K > 1$ and $L = 1$, the reporting label is Caution because alternative

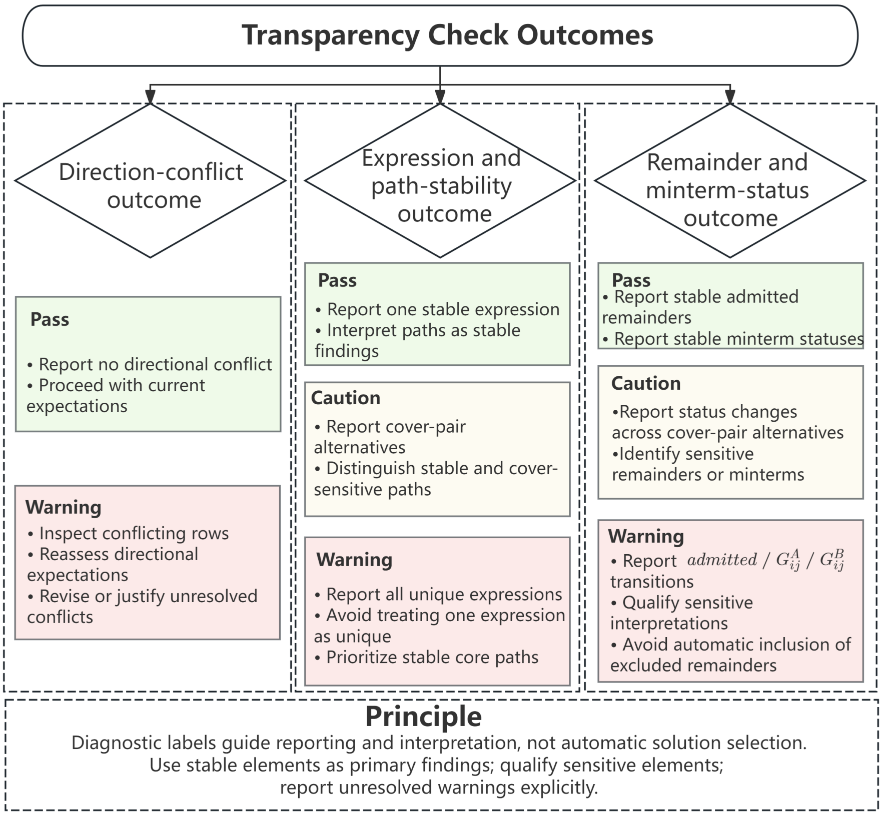

Fig. 3 Recommended reporting actions after transparency checks. The figure summarizes how Pass, Caution, and Warning labels can guide subsequent reporting and interpretation. These actions are intended to support transparent reporting rather than to automate solution selection or determine substantive validity.

cover-pair records exist but the final intermediate expression is unchanged. If $K > 1$ and $L > 1$, the reporting label is Warning because cover-pair records are associated with different intermediate expressions that should be disclosed.

Path stability is reported separately from full-expression stability. A stable core path appears under all cover-pair records. A cover-sensitive path appears under only some cover-pair records. This separation helps readers distinguish the cover-stable part of the intermediate result from paths that depend on the selected cover pair. After the admitted remainders are fixed, equivalent intermediate expressions may still be represented by different PI covers; such expression-level ambiguity is handled through the path-stability check and is conceptually distinct from remainder-level admission sensitivity. Stable core paths can receive primary reporting emphasis, whereas cover-sensitive paths should be interpreted with caution.

Table 2 Structural checks and reporting labels

Check	Label condition	Reporting implication
Direction conflict	$P\cap \downarrow_d N = \emptyset$: Pass	Report no conflict with observed negative rows.
	$P\cap \downarrow_d N \neq \emptyset$: Warning	Report conflicting rows and review directional coding.
Expression stability	$K = 1$: Pass	Report the single cover-based expression.
	$K > 1, L = 1$: Caution	Report cover-pair records and note the unchanged expression.
	$K > 1, L > 1$: Warning	Report unique expressions, stable core paths, and cover-sensitive paths.
Remainder/minterm status	All statuses stable: Pass	Report stable remainder/minterm statuses.
	Status changes, no admitted / non-admitted switch: Caution	Report status-sensitive rows.
	Admitted / non-admitted switch: Warning	Report admitted-remainder sensitivity and affected minterms.

4.3 Logical-Remainder and Minterm-Status Check

Each audit-relevant row is first distinguished as an observed positive row or a logical remainder. Logical remainders are then classified into four reporting statuses: admitted remainder, G^A_{ij}, G^B_{ij}, or not directionally reachable. This check operationalizes the admitted/G^A_{ij}/G^B_{ij} status switching introduced above by comparing whether logical remainders or minterms shift among these statuses across cover-pair records. Because logical remainders function as simplifying assumptions in parsimonious and intermediate solutions, changes in their admitted or excluded status are directly relevant to transparent interpretation (Ragin 2008; Schneider and Wagemann 2012). If all audit-relevant minterms are stable, the reporting label is Pass. If status changes occur but no admitted/non-admitted switch appears, the reporting label is Caution. If an admitted $\leftrightarrow$ non-admitted switch appears, the reporting label is Warning, where non-admitted includes G^A_{ij}, G^B_{ij}, and not directionally reachable. A switch between excluded categories, such as $G^A_{ij} \leftrightarrow G^B_{ij}$, is reportable but less consequential than an admitted/non-admitted switch. An admitted/non-admitted switch can qualify the substantive interpretation of which logical remainders support an intermediate expression.

Taken together, the three checks produce diagnostic reporting labels rather than statistical tests, validity scores, or automatic decisions about solution selection. This use of labels is consistent with robustness-oriented QCA reporting, where sensitivity information supports disclosure and interpretation rather than mechanically accepting or rejecting solutions (Oana and Schneider 2024). A Pass label indicates that the corresponding structure can be reported without extended qualification. A Caution label indicates reportable sensitivity that should be described but may not alter the final expression or admitted/non-admitted status. A Warning label indicates that the relevant source of sensitivity should be explicitly disclosed and incorporated into substantive interpretation. Figure 3 summarizes how these labels can guide reporting actions.

In reporting, stable elements can be given primary emphasis, whereas cover-sensitive paths and status-sensitive remainders should be explicitly qualified. Unresolved Warning labels should be reported directly rather than hidden behind a single final formula. These actions are used in the case demonstration below to organize the interpretation of cover-pair sensitivity and logical-remainder status changes.

5 Case Demonstration

5.1 Case Setting and Truth-Table Input

The case demonstration uses a school digital transformation fsQCA dataset constructed from 58 smart education sample schools. The dataset was prepared by the authors' research team and is used here as a methodological demonstration of the logical-remainder transparency report rather than as a study of the causal mechanisms of digital transformation.

The truth table is treated as the input. The prepared truth-table output is fixed, including a frequency cutoff of 1, a raw-consistency cutoff of 0.8, a PRI cutoff of 0.7, and the row classification used in the case dataset. The outcome is digital transformation level, denoted Y. The seven conditions are coded as A–G. Table 3 reports the variable names and directional expectations used in the transparency report. Conditions A–F are assigned positive directional expectations based on the case context and expert judgment. Condition G, home–school–enterprise–community collaboration, is coded as unspecified in the directional-expectation vector. This does not imply that G is substantively irrelevant; rather, the available case knowledge and expert judgment do not justify using G as a monotonic basis for directional counterfactual movement in this audit. The symbol $\circ$ therefore indicates that this dimension is not used to justify directional reachability in the methodological audit.

The complex-cover family exported for the audit contains one complex cover,

$$C_1 = \{\mathrm{ABCdeF}, \mathrm{AbCdefg}, \mathrm{AbcDeFG}\}. \tag{22}$$

5.2 Standard Intermediate-Solution Output for Comparison

The standard intermediate-solution output is used as the conventional point of comparison for the transparency report. It was generated in fsQCA 4.1 for Windows using the fixed truth-table input and the directional specification adopted in this audit. Under this specification, the selected intermediate expression is

$$\mathrm{ACg} + \mathrm{ABCF} + \mathrm{ADFG}. \tag{23}$$

Table 4 reproduces the corresponding solution formula and fit indices.

Table 3 Case variables and directional expectations

Symbol	Variable name	Meaning	Direction
Y	DTL	Digital transformation level	Outcome
A	PS	Policy support	1
B	PDS	Project-driven support	1
C	AEI	Award and evaluation incentives	1
D	OIS	Organizational and institutional support	1
E	ID	Infrastructure development	1
F	DRCU	Digital resource creation and use	1
G	HSEC	Home–school–enterprise–community collaboration	$\circ$

Table 4 Standard intermediate-solution output used for comparison

Path / solution	Raw coverage	Unique coverage	Consistency
$A * C * \sim G$ / ACg	0.248767	0.0690628	0.857837
$A * B * C * F$ / ABCF	0.272023	0.0591967	0.844639
$A * D * F * G$ / ADFG	0.183228	0.054616	0.761347
Solution	0.397815	–	0.812230

This output provides the formula-level summary normally reported in applied fsQCA studies. It identifies the selected intermediate expression and reports the associated coverage and consistency values. However, it does not show whether the expression is unique among cover-pair records, whether its component paths have the same degree of stability, or how directionally reachable logical remainders are admitted or excluded behind the reported expression.

To make the case demonstration interpretable, the three paths can be briefly described in substantive terms. The path ACg combines policy support and award/evaluation incentives with the absence of G. Because G is unspecified in the directional-expectation vector, this absence should be interpreted only as part of the observed path configuration, not as evidence that multi-actor collaboration has a negative effect. The path ABCF combines policy support, project-driven support, award/evaluation incentives, and digital resource creation and use, suggesting an external-catalyst and resource-creation logic. The path ADFG combines policy support, organizational and institutional support, digital resource creation and use, and multi-actor collaboration, suggesting a policy-guided organizational–resource–collaboration logic.

5.3 Transparency Report Inputs and Direction-Conflict Check

The specified direction vector is

$$d = (1, 1, 1, 1, 1, 1, \circ). \tag{24}$$

The truth-table partition is $|\Omega| = 128$, $|P| = 4$, $|N| = 8$, and $|R| = 116$. The direction-conflict check reports no conflicting positive rows under the specified vector.

5.4 Parsimonious-Cover Alternatives and Parsimonious Boundaries

Under the fixed audit inputs, the audit output enumerates one complex-cover alternative and 15 parsimonious-cover alternatives, yielding $K = 15$ cover-pair records. These records cover the same observed positive rows, but their parsimonious boundaries may differ. In this case, the parsimonious boundary sizes range from 44 to 92 remainders. This variation is one structural source of cover-pair sensitivity in the transparency report.

Supplementary Material S3 provides Figure S3.1, which visualizes the parsimonious-cover alternatives and the variation in their parsimonious-boundary sizes.

Table 5 Parsimonious-cover alternatives and parsimonious boundary sizes

ID	Parsimonious cover S_j	$\lvert B_j \rvert$
1	D + BC + Ag	92
2	D + Ag + CFG	88
3	D + Ag + ACF	84
4	BC + Ag + bFG	72
5	BC + Ag + bcF	68
6	BC + Ag + AbF	64
7	BC + Ag + Abc	64
8–15	Additional parsimonious-cover alternatives listed in Supplementary Material S3, Table S3.1	44–60

5.5 Path-Level Comparison: Standard Output and Transparency Report

The standard output places ACg, ABCF, and ADFG side by side as components of the selected intermediate expression. The transparency report qualifies this interpretation by showing that these paths do not have the same stability across cover-pair records.

Conditional on the fixed truth table, directional expectations, and minimization structure, the 15 cover-pair records are associated with $L = 6$ unique intermediate expressions. Under the reporting rules in Table 2, this receives a Warning label for reporting purposes because $K > 1$ and $L > 1$. The stable core path is ACg, which appears in all 15 records. By contrast, ABCF appears in 10 records and ADFG appears in only 3 records. The transparency report also reveals other cover-sensitive paths that do not appear in the selected standard output: AbDFG, AbcDFG, ABCdeF, and ABCeF.

Figure 4 reports the frequency with which each path appears across the 15 cover-pair records. Detailed cover-pair records, path-stability summaries, and minterm-status tables are provided in Supplementary Material S3.

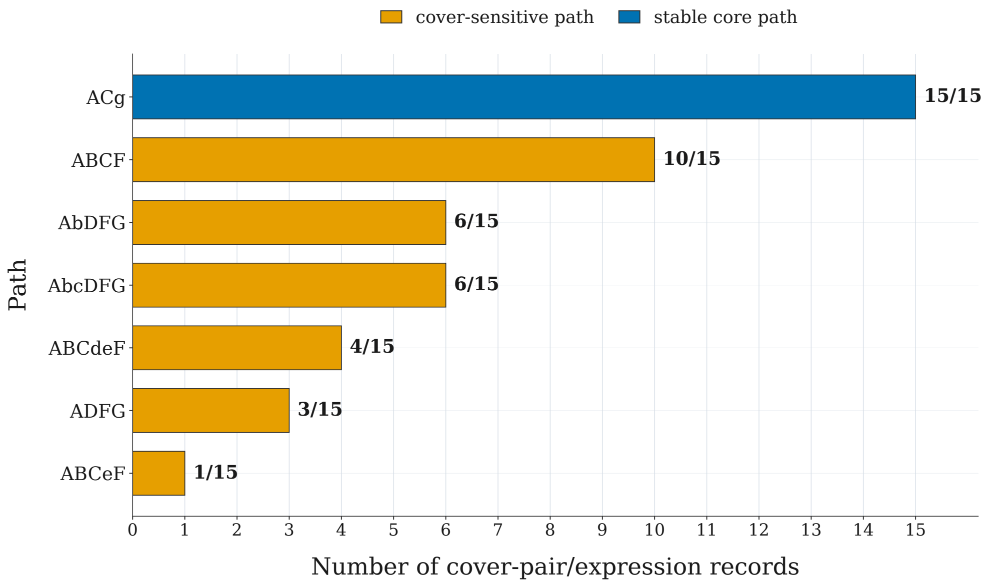

Fig. 4 Path stability across cover-pair records. The figure reports how often each intermediate path appears across the 15 cover-pair records. ACg appears in all records and is stable across covers, whereas the remaining paths appear only under some records and are cover-sensitive.

5.6 Remainder-Level Comparison: Standard Output and Transparency Report

The comparison is more detailed at the logical-remainder level. The standard output does not report which logical remainders are admitted by the standard term-pair procedure, which directionally reachable remainders remain excluded inside the selected boundary, or which are excluded outside that boundary. The transparency report identifies those objects explicitly.

Across all cover-pair records, the audit output records 281 admitted-remainder status entries, 21 G_{ij}^{A} entries, and 28 G_{ij}^{B} entries. These are status records over 15 cover-pair records rather than distinct remainders. At the remainder level, 15 remainders are admitted under all 15 records, 94 remainders are consistently not directionally reachable, and 7 show sensitive admitted-remainder status. Because admitted $\leftrightarrow G_{ij}^{A}/G_{ij}^{B}$

switches occur, the reporting label is Warning, meaning that the corresponding sensitivity should be disclosed.

Figure 5 displays the directionally reachable logical-remainder set $D = R \cap \uparrow_d P$ and its status profile across the cover-pair records. In general, $\overline{E} = \bigcup_{i,j} E_{ij} \subseteq D$. In this case, equality holds, meaning that every directionally reachable logical remainder is admitted in at least one cover-pair record, although its status may still vary across records.

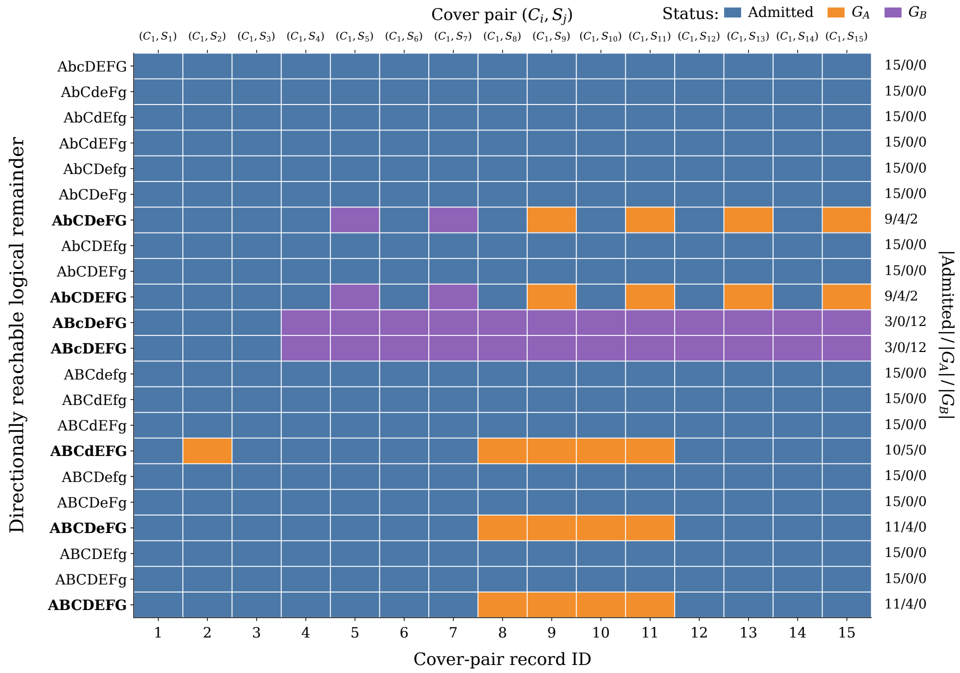

Fig. 5 Remainder-status profile matrix across cover-pair records. Each row records how one of the 22 directionally reachable logical remainders is classified across the cover-pair records. Blue indicates admitted remainders, orange indicates G^A_{ij}, and purple indicates G^B_{ij}. Rows with identical profiles indicate remainders that co-switch across the same cover-pair records, making status sensitivity visible at the remainder level.

Full minterm-stability counts are reported in Supplementary Material S3, Table S3.4. The seven sensitive remainders are not arbitrary Boolean rows. All of them contain $A = 1$, $F = 1$, and $G = 1$, placing them in a substantively meaningful region with policy support, digital resource creation and use, and multi-actor collaboration. They can be read in four case-informed groups. The first pair, $AbCDeFG$ and $AbCDEFG$, describes high-support configurations in which project-driven support B is absent, showing whether the absence of B is bridged by remainder admission under selected covers. The second pair, $ABcDeFG$ and $ABcDEFG$, describes configurations with policy support, project support, organizational support, digital resources, and collaboration, but without award/evaluation incentives C, showing whether incentive absence is treated differently across cover-pair records. The remainder $ABCdEFG$ represents a configuration in which external catalysts, resources, and collaboration are present, but organizational and institutional support D is absent; this case distinguishes directional reachability from standard term-pair admission. The final pair, $ABCDeFG$ and $ABCDEFG$, includes an infrastructure-absent configuration and the all-present configuration, showing that even substantively plausible remainders may have unstable admitted/G_{ij}^{A}/G_{ij}^{B} status.

These readings serve a methodological purpose. They show that the sensitive remainders are interpretable Boolean rows and that cover-pair records can redistribute such rows across admitted, within-boundary excluded, and outside-boundary excluded statuses. The transparency report therefore makes visible a counterfactual support structure that is not displayed in the standard intermediate-solution output.

5.7 Summary of the Case Demonstration

Without the transparency report, the case would likely be reported as the selected intermediate expression ACg+ABCF+ADFG, together with coverage and consistency values. The transparency report qualifies the same standard output by identifying one stable core path, several cover-sensitive paths, and seven remainders whose admitted/excluded status varies across cover-pair records. The gain is interpretive rather than algorithmic: the report clarifies what should be disclosed and how much interpretive weight should be placed on stable versus cover-sensitive components.

The standard intermediate-solution output and the transparency report therefore answer different reporting questions. The former reports the selected formula and fit indices; the latter converts cover-pair and remainder-status information into explicit reporting objects. These labels indicate that the final intermediate expression should be accompanied by disclosure of cover-pair records, stable and cover-sensitive paths, and remainder/minterm status.

Following the reporting logic summarized in Figure 3, the case results should be interpreted with different levels of emphasis. The direction-conflict check receives a Pass label, so the selected direction vector can be reported without extended conflict discussion. By contrast, the intermediate-expression check receives a Warning label because, under the fixed audit inputs, 15 cover-pair records are associated with six unique intermediate expressions. Therefore, the case should not be presented as if one intermediate expression were uniquely determined. The stable core path ACg can receive primary reporting emphasis, whereas the remaining paths should be treated as cover-sensitive.

The remainder/minterm-status check also receives a Warning label. Accordingly, the analysis should disclose the seven sensitive remainders and the admitted/G_{ij}^{A}/G_{ij}^{B} transitions across cover-pair records. These transitions show where the final expression depends on cover-pair selection and where substantive interpretation should be qualified. Table 6 summarizes these reporting implications.

Supplementary Material S4 applies the same transparency-report objects to the classical Lipset fuzzy-set example as a contrastive public-data check, illustrating the portability of the reporting framework across datasets. Unlike the school digital transformation case, the Lipset check produces a direction-conflict Warning under the

Table 6 Summary of the transparency report

Check	Label	Reporting implication
Direction conflict	Pass	The selected direction vector 111111∘ produces no conflicting positive rows
Intermediate expression stability	Warning	Under the fixed audit inputs, fifteen cover-pair records are associated with six unique intermediate expressions; report stable core and cover-sensitive paths
Logical remainder status stability	Warning	Seven remainders have sensitive admitted-remainder status; admitted, G^A_{ij}, and G^B_{ij} status should be reported

fixed all-positive directional vector, while cover-pair expression variation and minterm-status sensitivity are absent: $K = 1$, $L = 1$, both paths are stable within the single cover-pair record, $|G^A_{11}| = |G^B_{11}| = 0$, and no admitted/non-admitted switch appears. This contrast shows that the same reporting framework can distinguish different sources of reporting concern across cases rather than mechanically producing the same diagnostic pattern.

6 Discussion

6.1 From Formula-Level Output to Auditable Support Structures: Implications for Counterfactual Ordering

Standard QCA reporting conventionally foregrounds solution formulas and fit indices (Ragin 2008). In intermediate solutions, this output is shaped by directional expectations and the treatment of logical remainders (Schneider and Wagemann 2012). A formula-level expression compresses the row-level support structure from which the expression is derived: the final expression does not show how observed positive rows, admitted logical remainders, and excluded but directionally reachable remainders jointly shape the counterfactual support profile of a path. Limited diversity makes logical remainders central to QCA counterfactual reasoning (Ragin and Sonnett 2005), and careful treatment of remainders is therefore essential when intermediate solutions are interpreted (Schneider and Wagemann 2013).

The transparency report extends the interpretive unit from formula-level expressions to row-level support and boundary structures. The objects E_{ij}, G^A_{ij}, G^B_{ij}, path stability, and minterm-status stability make the support structure behind an intermediate expression auditable. They show whether a path is supported by admitted remainders, associated with unadmitted but directionally reachable remainders, or sensitive to the selected parsimonious boundary. The result is an interpretive shift that makes the counterfactual support, boundary position, and cover-pair sensitivity behind an intermediate expression available for inspection.

This audit layer also has a clear boundary. It keeps the standard intermediate-solution procedure in place and reports the sensitivity generated by the parsimonious boundary B_j, cover-pair selection (C_i, S_j), and term-pair admission $E(c, s)$. It makes these sensitivities visible and reportable, but it does not remove them algorithmically or generate a cover-independent solution. This boundary suggests a separate methodological question for future research: whether the order of counterfactual expansion and parsimonious minimization could be altered so that directional screening is performed before the minimization step that selects among parsimonious implicants.

Figure 6 organizes these relationships as a conceptual mapping of the standard procedure, the reporting layer developed here, and a possible future expand-first strategy. The third row is included to identify a future methodological direction whose formal and empirical properties remain to be evaluated.

$$
\begin{tikzcd}
(P,N,R,d) \arrow[r, mapsto, "\mathrm{QM}"] \arrow[ddrr, mapsto, rounded corners, to path={|- (\tikztotarget) \tikztonodes}, "d\text{-expansion}" near end] & (P,\mathcal{C},\mathcal{S},d) \arrow[r, mapsto, "\text{select}"] \arrow[d, mapsto, "\substack{\text{all pairs}\\ d\text{-expansion}\\ \text{boundary}}"] & (C_i,S_j,d) \arrow[r, mapsto, "d\text{-expansion}"] & (P,E_{ij}) \arrow[r, mapsto, "\mathrm{QM}"] & I_{ij} \\
 & \big(\{(P,E_{ij})\}_{i,j}, D, \{B_j\}_j\big) \arrow[rr, mapsto, "\mathrm{QM}\ \&\ \text{report}"] & & \big(\{I_{ij}\}_{i,j}, \{E_{ij}\}_{i,j}, \{G^A_{ij}, G^B_{ij}\}_{i,j}\big) & \\
 & & (P,E_d) \arrow[rr, mapsto, "\mathrm{QM}"] & & I^*
\end{tikzcd}
$$

Fig. 6 Conceptual mapping of standard intermediate-solution construction, transparency reporting, and a possible future ordering of counterfactual reasoning.

The first row of Figure 6 summarizes the standard construction discussed in Section 3.3. A selected cover pair (C_i, S_j) induces E_{ij}, and the corresponding intermediate expression is obtained by minimizing the expanded positive region, as formalized in Equation (15). Here $\mathcal{C}$ and $\mathcal{S}$ denote the complex-cover and parsimonious-cover families defined in Section 3.3.

The second row represents the transparency report developed in this article. It preserves the standard minimize-first, expand-second order while bringing all cover-pair records and their support-structure consequences into the reporting scope. The reporting objects include $\{I_{ij}\}_{i,j}$, $\{E_{ij}\}_{i,j}$, D, $\{B_j\}_j$, and $\{G^A_{ij}, G^B_{ij}\}_{i,j}$, making admitted remainders, excluded directionally reachable remainders, boundary variation, and cover-pair sensitivity inspectable.

The third row indicates a possible future methodological direction. Its distinguishing feature is that directional screening first produces a directionally expanded or admissible remainder set $E_d \subseteq R$, which is then supplied as restricted don't-care rows before minimization. As an illustrative starting point, E_d could initially be specified as $D = R \cap \uparrow_d P$, with further theoretical or conflict-screening restrictions imposed if needed. With such a restricted set supplied before minimization, the future expression can be written as:

$$I^* \in \mathrm{QM}_\Omega(P, E_d, \varnothing).$$

This differs from the standard I_{ij} construction, where admitted remainders have already been incorporated into the positive region before final minimization. Because E_d is specified before parsimonious-cover selection and term-pair admission, the minterm-level support of I^* would no longer be induced by B_j, (C_i, S_j), or $E(c, s)$. In this limited sense, an expand-first strategy would bypass the support-structure sensitivity that the transparency report reveals in the standard intermediate-solution procedure. This does not imply formula-level uniqueness, because equivalent or alternative minimized expressions may still arise; the claim is restricted to the minterm-level support structure determined by the fixed directional screening step. If E_d is uniquely determined by fixed P, R, and d, this strategy may therefore yield a minterm-level support structure that is less dependent on parsimonious boundaries, cover-pair records, and term-pair admission. The uniqueness, interpretability, equivalence properties, robustness, and practical validity of the resulting expression or expression family would require further formal and empirical validation.

6.2 From a Selected Cover to PI-Cover Multiplicity: Reallocating Interpretive Emphasis Across Paths and Remainders

The auditability issue discussed above becomes especially consequential when PI-cover multiplicity is present. Conventional reporting often foregrounds a selected or reported cover and expression; the transparency report asks how interpretation changes when the alternative cover-pair records behind that expression are also inspected. In the present framework, these consequences are traced through cover-pair records. Exact minimization work has shown that complete prime-implicant structures can be derived efficiently and should be inspected carefully (Dusa 2018). Model-ambiguity research further shows that alternative minimal models may arise from the same configurational information (Baumgartner and Thiem 2017). This concern resonates with broader warnings that model ambiguity and result determinacy require explicit attention in QCA applications (Thiem 2014). The transparency report extends this concern from the existence of alternatives to their consequences for interpretation. PI-cover multiplicity produces alternative technical representations and also reallocates the counterfactual support and interpretive emphasis attached to paths and remainders. Cover-pair alternatives should therefore be read as alternative minimization outcomes and as alternative openings of the Boolean space.

When cover-pair alternatives change B_j and E_{ij}, they also change which parts of the intermediate solution appear stable and which parts depend on cover selection. The case demonstration illustrates this point: the standard expression lists several paths side by side, whereas the transparency report differentiates ACg as a stable core path from paths such as ABCF and ADFG, whose appearance is cover-sensitive. Stable core paths can receive stronger interpretive emphasis, while cover-sensitive paths warrant explicit qualification.

Figure 5 provides a concrete remainder-level motivation for this summary. Each row is a directionally reachable logical remainder. A row that remains admitted across all cover-pair records corresponds to a remainder included in every E_{ij}, whereas a row that switches among admitted, G^A_{ij}, and G^B_{ij} indicates that its admission depends on the selected cover pair. The mixed-status rows in the case demonstration are therefore the empirical counterpart of cover-sensitive admitted remainders. This observation motivates a set-family summary of admission stability across all cover-pair records.

Across all complex-cover and parsimonious-cover alternatives, the admitted-remainder sets can be collected as the finite family $\mathcal{E} = \{E_{ij} : 1 \le i \le m,\ 1 \le j \le n\} \subseteq \mathcal{P}(R)$. Under set inclusion, this family has minterm-level lower and upper bounds, and the cover-sensitive part is the difference between them:

$$\underline{E} = \inf_{\subseteq} \mathcal{E} = \bigcap_{i,j} E_{ij}, \qquad \overline{E} = \sup_{\subseteq} \mathcal{E} = \bigcup_{i,j} E_{ij}, \qquad \Delta E = \overline{E} \setminus \underline{E}.$$

The set $\underline{E}$ identifies the stable admitted-remainder core, whereas $\overline{E}$ identifies the possible admitted-remainder set, namely remainders admitted under at least one cover pair. The difference ΔE marks cover-sensitive admitted remainders. For every cover-pair record, $\underline{E} \subseteq E_{ij} \subseteq \overline{E} \subseteq D$. Because $\mathcal{E}$ is finite, these bounds are uniquely determined by intersection and union. They therefore summarize minterm-level admission stability even when expression-level PI-cover ambiguity remains.

This notation also clarifies the connection with the expand-first strategy in Section 6.1. The set ΔE captures standard cover-pair admitted-remainder variation: the admissible-remainder input changes across cover-pair records. If an expand-first strategy instead supplies a single directionally screened admissible-remainder input E_d, the analogous family is $\mathcal{E}^* = \{E_d\}$, so $\underline{E}^* = \overline{E}^* = E_d$ and $\Delta E^* = \varnothing$. The potential advantage is the removal of admissible-remainder input-level variation, not a guarantee of formula-level uniqueness after minimization, because equivalent or alternative minimized expressions may still arise.

The same set-family perspective points to a more conservative counterpart. Rather than moving admission before cover-pair selection, a stable-core expression would retain the standard cover-pair procedure and minimize only the admitted-remainder core shared by all cover-pair records:

$$\underline{I} = \mathrm{QM}_{\Omega}(P \cup \underline{E}, \varnothing, \varnothing).$$

The input support $P \cup \underline{E}$ is uniquely determined at the minterm level because $\underline{E}$ is the intersection of all E_{ij}. Unlike I^*, however, $\underline{I}$ does not bypass the standard cover-pair procedure; it conservatively summarizes the common support that survives across that procedure. It should therefore be read as a possible future extension, not as an additional output of the present reporting framework or a validated alternative algorithm. Its usefulness as a stable-core expression would require further assessment, especially with respect to coverage loss, interpretability, robustness, and its relationship to standard intermediate expressions.

For interpretation, the main implication remains at the remainder and minterm level. Admitted/G^A_{ij}/G^B_{ij} switching indicates that some counterfactual support is not fixed by directional expectations alone, but by the interaction between those expectations and the selected cover pair. Cover alternatives can therefore redistribute interpretive emphasis across paths and logical remainders even under fixed upstream inputs. A transparency report changes the evidentiary reading of the intermediate solution by separating stable support from cover-dependent support and by making the latter visible before substantive interpretation is drawn from the final expression.

6.3 Limitations and Future Research

The framework should be read within clear scope conditions: it starts after calibration, threshold selection, truth-table construction, and directional specification have already been made. It assumes a fixed truth table, fixed directional expectations, and enumerated complex-cover and parsimonious-cover alternatives. These scope conditions align with calls to treat QCA inference as a research-design problem involving validity claims and explicit modes of reasoning (Thomann and Maggetti 2020). The framework improves the inspectability of intermediate-solution reporting, but it does not validate the substantive correctness of calibration decisions, thresholds, or directional expectations. The direction-conflict check can reveal a structural inconsistency between directional expectations and observed negative rows, but it does not determine whether the directional expectations are theoretically justified.

The empirical demonstration should be read in the same way. The school digital transformation case demonstrates how the report works in a substantive setting, while Supplementary Material S4 provides a Classical Lipset public-case check for portability. This distinction matters because the framework improves the inspectability of one specific stage of QCA reporting, while broader debates about QCA validity, model ambiguity, and measurement error remain outside its scope (Baumgartner and Thiem 2020).

Future research can move from demonstration to validation in three directions. First, the report should be applied to a wider range of public datasets, software environments, truth-table structures, boundary cases, and versioned implementation releases. Second, future work should examine when support-structure instability is most likely to arise, including whether it is associated with limited diversity, calibration choices, threshold settings, PI-chart structure, or the number of cover-pair alternatives. Third, future work should assess the methodological extensions suggested by the discussion above. One direction is the expand-first strategy, which supplies a directionally screened admissible-remainder set before minimization and should be evaluated for its ability to reduce cover-pair-induced input variation without weakening QCA's interpretive logic. A second, more conservative direction is the stable-core expression $\underline{I}$, which should be assessed in terms of coverage loss, interpretability, robustness, and its relationship to standard intermediate expressions. These analyses could then assess whether support-structure instability can be reduced through more theory-constrained directional expectations, more transparent calibration anchors, sensitivity-informed threshold choices, or principled procedures for handling PI-cover multiplicity.

7 Conclusion

FsQCA intermediate solutions should not be reported only as final formulas accompanied by fit indices. Because intermediate solutions depend on directional expectations and logical remainders, their interpretation also depends on a counterfactual support structure that is usually compressed in conventional output. The proposed transparency report makes this structure inspectable by distinguishing admitted remainders, within-boundary excluded remainders, and outside-boundary excluded remainders, and by tracing how paths and minterm statuses behave across cover-pair records.

The case demonstration shows why this additional reporting layer matters. The selected intermediate expression provides a useful formula-level summary, but it does not reveal that one path is stable across all cover-pair records whereas other paths are cover-sensitive, nor does it show that some directionally reachable remainders shift between admitted and excluded statuses. The framework therefore does not replace standard fsQCA minimization or select a preferred solution automatically. Instead, it clarifies which parts of an intermediate solution can receive primary interpretive emphasis and which parts require qualification or disclosure.

The contribution is therefore primarily a reporting and interpretive one, but it also has a broader methodological implication. By turning the counterfactual support behind intermediate solutions into auditable objects, the framework helps applied researchers make the use of logical remainders more transparent, reproducible, and open to scrutiny while preserving the standard minimization procedure. At the same time, the observed dependence of support structures on parsimonious boundaries, cover-pair records, and term-pair admission points to a future agenda: assessing whether expand-first strategies and conservative stable-core expressions can reduce support-structure instability while preserving QCA's interpretive logic.

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Author Contributions

The authors declare the following contributions according to the CRediT taxonomy. Jiahao Miao: Conceptualization, Methodology, Formal Analysis, Software, Visualization, Writing–Original Draft, and Writing–Review & Editing. Yi Tao: Data Curation, Investigation, and Validation. Xiao-Fan Lin: Project Administration, Supervision, and Writing–Review & Editing. All authors reviewed and approved the final manuscript.

Code and Data Availability

The supporting code is publicly available at https://github.com/AbelMiao666/fsqca-intermediate-solution-audit. The fixed truth-table input used in the school digital transformation case demonstration is provided as the supplementary data file `case1_true_table.csv`. The fixed truth-table input used in the Classical Lipset fuzzy-set public-case check is provided as the supplementary data file `case2_true_table.csv`. The complete audit outputs generated by the supporting implementation are provided as `case1_fsqca_logical_remainder_audit_report.xlsx` and `case2_fsqca_logical_remainder_audit_report.xlsx`. Full pseudocode, input/output descriptions, implementation notes, full case-output descriptions, and the additional public-case check are provided in the Supplementary Materials.

Appendix A Procedural Rules and Proof of the Term-Pair Characterization

The seven procedural rules summarized in Table A1 use Ragin and Sonnett (2005)'s account of limited diversity and counterfactual cases as the substantive background for understanding logical remainders. They also draw on standard QCA accounts of intermediate minimization and directional expectations (Ragin 2008; Schneider and Wagemann 2012), as well as Dusa (2019)'s implementation-oriented treatment of truth-table construction, logical minimization, and pseudo-counterfactual analysis in applied workflows.

Building on this literature, the direct procedural basis for the equivalence proved in Appendix A is Mendel and Korjani (2012)'s formal reconstruction of Ragin's fsQCA intermediate-solution procedure. Appendix A.1 translates that formalized intermediate-solution computation process into the notation used here. Appendix A.2 shows that, under fixed truth-table inputs, fixed directional expectations, and a fixed cover pair (C_i, S_j), the local rule-based acceptable-remainder set in that procedure coincides with the term-pair extension set $E(c, s)$. Thus, the result establishes a procedural equivalence between the formalized fsQCA intermediate-solution computation and the row-level reporting object, rather than proposing a new minimization rule.

Appendix A.1 Seven procedural rules behind term-pair admission

For a candidate logical remainder $r \in R$, the rules in Table A1 reduce to three row-level filters under an effective complex–parsimonious term pair (c, s): effective-pair membership $(c, s) \in \Lambda_{ij}$, directional reachability from the complex source region $r \in \uparrow_d (\Gamma(c))$, and local boundary membership $r \in \Gamma(s)$. Hence, for each effective pair (c, s), the corresponding local term-pair extension set is

$$E(c, s) = R \cap \Gamma(s) \cap \uparrow_d (\Gamma(c)).$$

Appendix A.2 Proof of Proposition 1

The proof uses the following equivalent statement of Proposition 1. Conditional on a fixed truth-table partition (P, N, R), a fixed directional-expectation vector d, and a fixed cover pair (C_i, S_j), the local rule-based acceptable-remainder set and the term-pair extension set coincide:

$$A(c, s) = E(c, s) = R \cap \Gamma(s) \cap \uparrow_d (\Gamma(c)) \quad \text{for every } (c, s) \in \Lambda_{ij}.$$

Proof. Let

$$\Lambda_{ij} = \{\, (c, s) \in C_i \times S_j : \Gamma(c) \subseteq \Gamma(s) \,\}.$$

For $x, y \in \Omega$, recall that $x \preceq_d y$ means that every changed coordinate moves only to the value specified by d_j, while no movement is allowed when $d_j = \circ$. Hence,

$$r \in \uparrow_d (\Gamma(c)) \iff \exists x \in \Gamma(c) \text{ such that } x \preceq_d r.$$

Table A1 Seven procedural rules behind term-pair admission

Rule	Rule stated in the notation used here
R1	Counterfactual analysis is performed only for an effective complex–parsimonious term pair. Here, this corresponds to $(c,s) \in \Lambda_{ij}$, equivalently $\Gamma(c) \subseteq \Gamma(s)$.
R2	The complex-solution term acts as the initial counterfactual source. Here, the source region is $\Gamma(c)$, with $\Gamma(c) \subseteq P$.
R3	If the directional expectation is unspecified for condition j, no directional movement is allowed on that dimension. Here, if $d_j = \circ$, then a candidate remainder must satisfy $r_j = x_j$ for the source minterm $x \in \Gamma(c)$.
R4	If the candidate remainder and the source minterm already agree on condition j, no change is made on that coordinate. This is automatically allowed by $x \preceq_d r$.
R5	If a condition is not explicit in the current counterfactual term, absorption may leave the counterfactual unchanged. In the set-theoretic characterization, this effect is handled through the selected complex term, the selected parsimonious boundary, and the union over effective term pairs; it is not added as a separate local filter.
R6	If a candidate remainder differs from the source minterm on condition j, the movement is admissible only when the change follows the stated directional expectation. Here, this means $d_j \in \{0,1\}$ and $r_j = d_j$.
R7	A directionally admissible candidate remainder is admitted only if it remains inside the local parsimonious boundary selected by the parsimonious term. Here, this corresponds to $r \in \Gamma(s)$.

First, take any $r \in A(c,s)$. By the procedural rules, $r \in R$, $(c,s) \in \Lambda_{ij}$, and $r \in \Gamma(s)$. The same rules require a source minterm $x \in \Gamma(c)$ such that movement from x to r follows the directional expectation vector d; equivalently, $x \preceq_d r$. Therefore $r \in \uparrow_d (\Gamma(c))$, and thus

$$r \in R \cap \Gamma(s) \cap \uparrow_d (\Gamma(c)) = E(c,s).$$

This proves $A(c,s) \subseteq E(c,s)$.

Second, take any $r \in E(c,s)$. Then $r \in R$, $r \in \Gamma(s)$, and $r \in \uparrow_d (\Gamma(c))$. By the definition of the upward directional closure, there exists $x \in \Gamma(c)$ such that $x \preceq_d r$. This satisfies the source-row requirement, the directional-movement requirements, and the local boundary requirement in Rules R2–R7. Because $(c,s) \in \Lambda_{ij}$ is fixed, Rule R1 also holds. Hence $r \in A(c,s)$, proving $E(c,s) \subseteq A(c,s)$.

The two inclusions give $A(c,s) = E(c,s)$. Taking the union over all effective pairs yields

$$E_{ij} = \bigcup_{(c,s)\in\Lambda_{ij}} E(c,s) = \bigcup_{c\in C_i,\ s\in S_j,\ \Gamma(c)\subseteq\Gamma(s)} E(c,s),$$

and adding the positive rows gives the row-level support $P \cup E_{ij}$, which is the expanded positive region minimized in Equation (15). $\square$

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Supplementary Materials for

“Opening the Black Box of Logical Remainders: A Transparency Reporting Framework for fsQCA Intermediate Solutions”

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S1 Formal Properties and Proofs

The propositions in Supplementary Material S1 establish the set-theoretic properties of the reporting objects used by the framework, including directional reachability, cover-pair containment, status partitions, and gap decompositions. The pseudocode in Supplementary Material S2 operationalizes these objects for reproducible reporting, while the propositions clarify why the resulting categories are mutually defined, bounded, and interpretable as ex post reporting categories within the standard output structure.

S1.1 Truth-table row-level formalization

The logical-remainder transparency report starts after calibration, threshold selection, and truth-table construction have already been completed. The formal objects in this supplement are therefore truth-table row-level objects. Let $\Omega = \{0,1\}^q$ denote the Boolean space of q conditions. A fixed truth-table partition is written as

$$\Omega = P \sqcup N \sqcup R,$$

where P is the set of positive rows, N is the set of observed negative rows, and R is the set of logical remainders. For a Boolean term t, $\Gamma(t) \subseteq \Omega$ denotes the set of rows covered by t. For a set of terms S, $\Gamma(S) = \bigcup_{s \in S} \Gamma(s)$.

The framework takes calibration, frequency thresholds, consistency and PRI cutoffs, truth-table construction, and theoretical cover selection as prior analytic decisions. Its inputs are a fixed truth-table partition, a fixed directional-expectation vector, enumerated complex-cover alternatives, and enumerated parsimonious-cover alternatives for ex post reporting.

S1.2 Directional reachability and conflict checking

Let $d = (d_1, \ldots, d_q) \in \{0, 1, \circ\}^q$ be the directional-expectation vector. For $x, y \in \Omega$, define

$$x \preceq_d y \quad \Longleftrightarrow \quad \text{for every } j,\ x_j \neq y_j \text{ implies } d_j \in \{0,1\} \text{ and } y_j = d_j.$$

If $d_j = \circ$, movement on dimension j is not justified by the directional expectation. For $A \subseteq \Omega$, define the directional upward closure and downward closure as

$$\uparrow_d A = \{y \in \Omega : \exists x \in A,\ x \preceq_d y\}, \qquad \downarrow_d A = \{x \in \Omega : \exists y \in A,\ x \preceq_d y\}.$$

Proposition S1. Directional reachability is a partial order: the relation $\preceq_d$ is reflexive, antisymmetric, and transitive on Ω.

Proof. Reflexivity holds because $x_j = y_j$ for all j when $x = y$. Antisymmetry holds because if $x \preceq_d y$ and $y \preceq_d x$, then any changed coordinate would have to end at the same specified directional value in both directions, which is impossible unless the coordinate is unchanged. Hence $x = y$. For transitivity, suppose $x \preceq_d y$ and $y \preceq_d z$. If

$x_j \neq z_j$, then either $y_j \neq z_j$, in which case $z_j = d_j$, or $y_j = z_j$, in which case $x_j \neq y_j$ and $y_j = d_j = z_j$. Thus $x \preceq_d z$. □

Proposition S2. Monotonicity of directional closures: if $A \subseteq B \subseteq \Omega$, then

$$\uparrow_d A \subseteq \uparrow_d B, \qquad \downarrow_d A \subseteq \downarrow_d B.$$

Proof. If $y \in \uparrow_d A$, then $x \preceq_d y$ for some $x \in A$. Since $A \subseteq B$, the same x is also in B, so $y \in \uparrow_d B$. The proof for downward closures is identical: if $x \in \downarrow_d A$, then $x \preceq_d y$ for some $y \in A \subseteq B$, hence $x \in \downarrow_d B$. □

Proposition S3. Equivalent forms of the direction-conflict check:

$$P \cap \downarrow_d N = \varnothing \quad \iff \quad \uparrow_d P \cap N = \varnothing.$$

Proof. If $P \cap \downarrow_d N \neq \varnothing$, then some $p \in P$ reaches some $n \in N$, so $n \in \uparrow_d P \cap N$. Conversely, if $\uparrow_d P \cap N \neq \varnothing$, then some $n \in N$ is reachable from some $p \in P$, so $p \in P \cap \downarrow_d N$. Thus the two emptiness conditions are equivalent. This equivalence shows that the direction-conflict check can be read as the absence of observed negative rows in the directional reachable region of the positive rows. It is a structural screen for reporting, not a test of theoretical correctness. □

S1.3 Reporting-region relations

The procedural-rule mapping and proof of the term-pair characterization are retained in Appendix A of the main article. Let

$$\mathcal{C} = \{C_1, \ldots, C_m\}, \qquad \mathcal{S} = \{S_1, \ldots, S_n\},$$

where each C_i is a complex-cover alternative and each S_j is a parsimonious-cover alternative. For each cover pair, the effective term-pair set is

$$\Lambda_{ij} = \{(c, s) \in C_i \times S_j : \Gamma(c) \subseteq \Gamma(s)\}.$$

The term-pair extension set is

$$E(c, s) = R \cap \Gamma(s) \cap \uparrow_d (\Gamma(c)),$$

and the cover-pair admitted-remainder set is

$$E_{ij} = \bigcup_{(c,s) \in \Lambda_{ij}} E(c, s).$$

Let

$$B_j = R \cap \Gamma(S_j), \qquad D = R \cap \uparrow_d P.$$

B_j is the logical-remainder portion of the selected parsimonious boundary, and D is the common directionally reachable logical-remainder set. Because every complex-cover alternative covers the positive rows, $\Gamma(C_i) = P$, this set does not vary with i:

$$R\cap \uparrow_d (\Gamma(C_i)) = R\cap \uparrow_d P = D.$$

Proposition S4. Cover-pair containment: for every cover pair (C_i, S_j),

$$E_{ij} \subseteq B_j \cap D \subseteq D.$$

Proof. If $r \in E_{ij}$, then $r \in E(c, s)$ for some $(c, s) \in \Lambda_{ij}$. Hence $r \in R$, $r \in \Gamma(s) \subseteq \Gamma(S_j)$, and $r \in\uparrow_d (\Gamma(c))$. Since $c \in C_i$ and $\Gamma(C_i) = P$, $\Gamma(c) \subseteq P$, so $r \in\uparrow_d P$. Thus $r \in B_j \cap D$. The inclusion $B_j \cap D \subseteq D$ is immediate. □

Proposition S5. Boundary monotonicity: if $B_{j_1} \subseteq B_{j_2}$, then

$$B_{j_1} \cap D \subseteq B_{j_2} \cap D.$$

Proof. If $r \in B_{j_1} \cap D$, then $r \in B_{j_1}$ and $r \in D$. Since $B_{j_1} \subseteq B_{j_2}$, it follows that $r \in B_{j_2} \cap D$. This monotonicity applies to the boundary-restricted directionally reachable region. It does not imply $E_{ij_1} \subseteq E_{ij_2}$, because admitted remainders also depend on the effective term-pair structure Λ_{ij}. □

S1.4 Remainder-status partition and equivalence conditions

The two excluded-remainder reporting categories are

$$G^A_{ij} = (B_j \cap D) \setminus E_{ij}, \qquad G^B_{ij} = D \setminus B_j.$$

G^A_{ij} records within-boundary excluded remainders; G^B_{ij} records outside-boundary excluded remainders. Both are ex post reporting categories rather than inclusion rules or substantive causal claims.

Proposition S6. Remainder-status decomposition: for a fixed cover pair (C_i, S_j),

$$B_j \cap D = E_{ij} \cup G^A_{ij}, \qquad D = E_{ij} \cup G^A_{ij} \cup G^B_{ij}.$$

Moreover, the logical remainders in R are partitioned into four mutually exclusive categories:

$$E_{ij}, \quad G^A_{ij}, \quad G^B_{ij}, \quad R \setminus D.$$

Proof. The first equality follows directly from $G^A_{ij} = (B_j \cap D) \setminus E_{ij}$ and $E_{ij} \subseteq B_j \cap D$. The second follows by adding $G^B_{ij} = D \setminus B_j$ and using $B_j \cap D \subseteq D$. The three sets E_{ij}, G^A_{ij}, and G^B_{ij} are mutually disjoint by construction and their union is D. The remaining logical remainders are outside D, meaning that they are not directionally reachable from P. Hence the four sets are disjoint and their union is R. This partition assigns reporting statuses rather than causal classifications. □

Proposition S7. Equivalence conditions for empty reporting gaps:

$$G^A_{ij} = \varnothing \iff E_{ij} = B_j \cap D,$$

$$G_{ij}^B = \varnothing \iff B_j \cap D = D,$$
$$G_{ij}^A = \varnothing \text{ and } G_{ij}^B = \varnothing \iff E_{ij} = D.$$

Proof. The first equivalence follows from $G_{ij}^A = (B_j \cap D) \setminus E_{ij}$ and $E_{ij} \subseteq B_j \cap D$. The second follows from $G_{ij}^B = D \setminus B_j$, which is empty exactly when all elements of D lie inside B_j. For the third equivalence, if both gaps are empty, then $E_{ij} = B_j \cap D = D$. Conversely, if $E_{ij} = D$, then $E_{ij} = B_j \cap D$ and $B_j \cap D = D$, so both gaps are empty. These equivalences mean that the corresponding excluded-remainder gap is absent from the transparency report; they do not imply that a cover pair is substantively preferable. □

S1.5 Infimum and supremum of the admitted-remainder family

Let
$$\mathcal{E} = \{E_{ij} : 1 \le i \le m,\ 1 \le j \le n\} \subseteq \mathcal{P}(R).$$
Define
$$\underline{E} = \inf_{\subseteq} \mathcal{E} = \bigcap_{i=1}^{m} \bigcap_{j=1}^{n} E_{ij},$$
$$\overline{E} = \sup_{\subseteq} \mathcal{E} = \bigcup_{i=1}^{m} \bigcup_{j=1}^{n} E_{ij},$$
$$\Delta E = \overline{E} \setminus \underline{E}.$$

Proposition S8. Infimum and supremum of the admitted-remainder family:
$$\underline{E} \subseteq E_{ij} \subseteq \overline{E} \subseteq D, \qquad \forall i, j.$$

Moreover, $\underline{E}$ and $\overline{E}$ are uniquely determined as the meet and join of $\mathcal{E}$ under set inclusion.

Proof. The intersection $\bigcap_{i,j} E_{ij}$ is contained in every E_{ij}. If a set X is contained in every E_{ij}, then $X \subseteq \bigcap_{i,j} E_{ij}$; hence the intersection is the greatest lower bound of $\mathcal{E}$ under set inclusion. The union $\bigcup_{i,j} E_{ij}$ contains every E_{ij}. If a set Y contains every E_{ij}, then $\bigcup_{i,j} E_{ij} \subseteq Y$; hence the union is the least upper bound of $\mathcal{E}$. By Proposition S4, $E_{ij} \subseteq D$ for all i, j, so $\overline{E} \subseteq D$. Since $\mathcal{E}$ is a finite family of subsets of R, these bounds are uniquely determined at the remainder/minterm coverage level. This is not an expression-level uniqueness result and does not select among cover-pair records. □

Proposition S9. Membership characterization of stable and cover-sensitive admitted remainders:
$$r \in \underline{E} \iff \forall i, j,\ r \in E_{ij},$$
$$r \in \overline{E} \iff \exists i, j,\ r \in E_{ij},$$
$$r \in \Delta E \iff (\exists i, j,\ r \in E_{ij}) \text{ and } (\exists i', j',\ r \notin E_{i'j'}).$$

Proof. The first two equivalences follow from the definitions of intersection and union. For $\Delta E = \overline{E} \setminus \underline{E}$, membership requires membership in at least one admitted-remainder set and absence from at least one admitted-remainder set. Thus $\underline{E}$ identifies the stable admitted-remainder core, $\overline{E}$ identifies the possible admitted-remainder set, and ΔE identifies cover-sensitive admitted remainders. □

S2 Pseudocode and Supporting Implementation

S2.1 Pseudocode for the logical-remainder transparency report

Table S2.1 summarizes the input and output objects used by the supporting implementation.

Table S2.1 Input and output objects in the supporting implementation

Object	Type	Role
P, N, R	Input	Fixed truth-table partition used by the report
d	Input	Directional-expectation vector used to compute reachability
$\mathcal{C}$	Input	Complex-cover alternatives $\{C_1, \dots, C_m\}$
$\mathcal{S}$	Input	Parsimonious-cover alternatives $\{S_1, \dots, S_n\}$
$E(c, s)$, E_{ij}	Output	Term-pair extension sets and cover-pair admitted-remainder sets
G_{ij}^A, G_{ij}^B	Output	Ex post reporting categories for within-boundary and outside-boundary excluded remainders
Path records	Output	Stable core and cover-sensitive paths across cover-pair records
Status matrix	Output	Minterm and remainder statuses across cover-pair records
Reporting labels	Output	Pass, Caution, or Warning labels for transparent reporting

The following pseudocode summarizes the supporting implementation at the level of reporting objects. Together, Algorithm S1 and Algorithm S2 make the reporting layer reproducible and inspectable at two levels of the report: cover-pair construction and cross-record comparison.

Algorithm S1 constructs the reporting objects for a single cover pair. It computes the admitted-remainder set, the within-boundary and outside-boundary excluded-remainder categories, and the per-record minterm, expression, and path records. The helper operations in Algorithm S1 abbreviate definitions from the main article and Supplementary Material S1:

- ClassifyMinterms($\cdot$) assigns each row or logical remainder to the reporting-status categories defined in the main article: positive row, observed negative row, admitted remainder, G_{ij}^A, G_{ij}^B, or not directionally reachable.
- RecordExpressionAndPaths($\cdot$) records the intermediate expression generated by minimizing the expanded positive region $P \cup E_{ij}$, as stated in Equation (15) of the main article, and records its component paths.

Algorithm S1: Per-cover-pair construction of reporting objects

Require: (P, N, R); d; one complex cover C_i; one parsimonious cover S_j

Ensure: Cover-pair reporting record Rec_{ij}

$B_j \leftarrow R \cap \Gamma(S_j)$

$D \leftarrow R \cap \uparrow_d P$

$\Lambda_{ij} \leftarrow \{(c, s) : c \in C_i,\ s \in S_j,\ \Gamma(c) \subseteq \Gamma(s)\}$

$E_{ij} \leftarrow \varnothing$

foreach $(c, s) \in \Lambda_{ij}$ **do**

  $E(c, s) \leftarrow R \cap \Gamma(s) \cap \uparrow_d (\Gamma(c))$

  $E_{ij} \leftarrow E_{ij} \cup E(c, s)$

$G^A_{ij} \leftarrow (B_j \cap D) \setminus E_{ij}$

$G^B_{ij} \leftarrow D \setminus B_j$

$\mathrm{Status}(C_i, S_j) \leftarrow \mathrm{ClassifyMinterms}(P, N, R, D, E_{ij}, G^A_{ij}, G^B_{ij})$

$(\mathrm{Expression}(C_i, S_j), \mathrm{PathRecord}(C_i, S_j)) \leftarrow$ $\mathrm{RecordExpressionAndPaths}(P \cup E_{ij})$

$\mathrm{Rec}_{ij} \leftarrow \{E_{ij}, B_j, D, G^A_{ij}, G^B_{ij},$
$\mathrm{Status}(C_i, S_j), \mathrm{Expression}(C_i, S_j),$
$\mathrm{PathRecord}(C_i, S_j)\}$

Return Rec_{ij}

Algorithm S2 compares the cover-pair records and assigns the direction-conflict, expression/path-stability, and remainder/minterm-status labels. The helper operations in Algorithm S2 summarize these cover-pair records:

- PerCoverPairReport$(\cdot)$ denotes the cover-pair construction procedure in Algorithm S1 and returns Rec_{ij}.
- UniqueExpressions$(\cdot)$ extracts and deduplicates intermediate expressions across cover-pair records.
- ComparePaths$(\cdot)$ records which paths appear under which cover-pair records.
- SplitPaths$(\cdot)$ separates paths that appear in all cover-pair records from cover-sensitive paths.
- CompareStatuses$(\cdot)$ compares minterm and logical-remainder statuses across cover-pair records.
- SensitiveMinterms$(\cdot)$ identifies minterms whose reporting status changes, especially admitted/non-admitted switches.
- LabelDirectionConflict$(\cdot)$, LabelPathStability$(\cdot)$, and LabelStatusStability$(\cdot)$ assign the three reporting labels according to the rules in the main article.

These helper operations operate at the reporting stage.

Algorithm S2: Cross-cover-pair comparison and reporting labels

Require: (P, N, R); d; $\mathcal{C} = \{C_1, \ldots, C_m\}$; $\mathcal{S} = \{S_1, \ldots, S_n\}$

Ensure: ℓ_{dir}, ℓ_{path}, ℓ_{status}, logical-remainder transparency report

$C_{\text{dir}} \leftarrow P \cap \downarrow_d N$

$\ell_{\text{dir}} \leftarrow \text{LabelDirectionConflict}(C_{\text{dir}})$

foreach $C_i \in \mathcal{C}$ **do**

  foreach $S_j \in \mathcal{S}$ **do**

    $\text{Rec}_{ij} \leftarrow \text{PerCoverPairReport}(P, N, R, d, C_i, S_j)$

$\mathcal{X} \leftarrow \text{UniqueExpressions}(\{\text{Rec}_{ij} : 1 \leq i \leq m,\ 1 \leq j \leq n\})$

$\text{PathSummary} \leftarrow \text{ComparePaths}(\{\text{PathRecord}(C_i, S_j) : 1 \leq i \leq m,\ 1 \leq j \leq n\})$

$(\mathcal{P}_{\text{stable}}, \mathcal{P}_{\text{sensitive}}) \leftarrow \text{SplitPaths}(\text{PathSummary})$

$\text{StatusSummary} \leftarrow \text{CompareStatuses}(\{\text{Status}(C_i, S_j) : 1 \leq i \leq m,\ 1 \leq j \leq n\})$

$\mathcal{M}_{\text{sensitive}} \leftarrow \text{SensitiveMinterms}(\text{StatusSummary})$

$\ell_{\text{path}} \leftarrow \text{LabelPathStability}(\mathcal{X}, \text{PathSummary})$

$\ell_{\text{status}} \leftarrow \text{LabelStatusStability}(\text{StatusSummary})$

Return $\text{TransparencyReport}(\ell_{\text{dir}}, \ell_{\text{path}}, \ell_{\text{status}}, \mathcal{X}, \mathcal{P}_{\text{stable}}, \mathcal{P}_{\text{sensitive}}, \mathcal{M}_{\text{sensitive}})$

S2.2 Implementation note

The supporting implementation was written in Python 3.12 and uses a fixed truth-table input and fixed minimization-related settings to generate the reporting objects used in the case demonstration. Frequency, consistency, and PRI thresholds are treated as user-specified analytic settings. Calibration, truth-table construction from raw case-level data, and final theoretical cover selection remain prior analytic steps. The implementation makes the downstream remainder-admission and remainder-exclusion structure inspectable after the relevant inputs and analytic settings have been fixed.

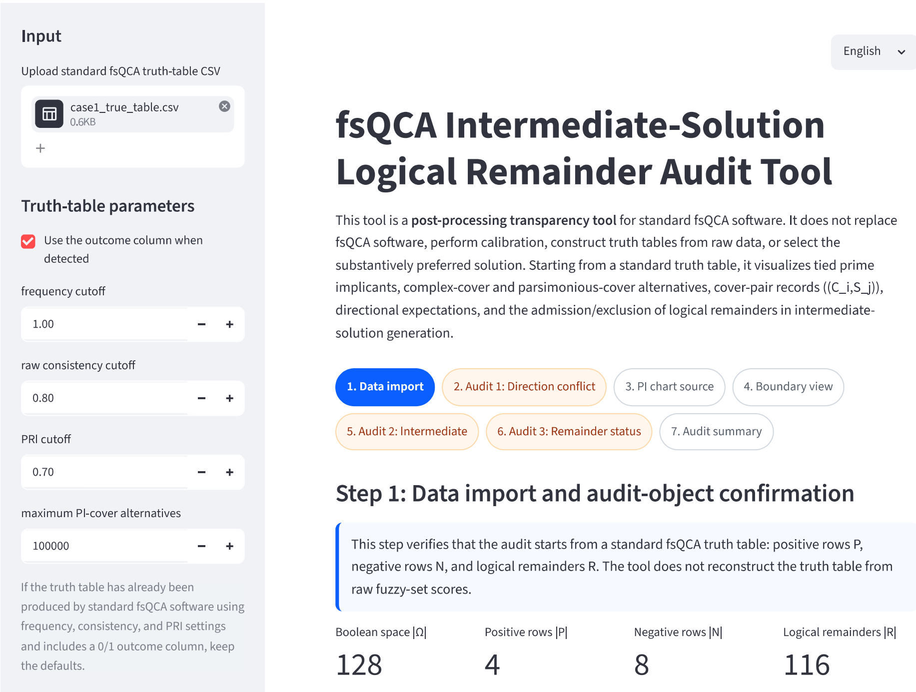

Fig. S2.1 Example report produced by the supporting implementation. The report records fixed truth-table inputs, cover alternatives, intermediate-expression and path-stability results, and admitted/excluded logical-remainder statuses.

S3 Full Case Output

S3.1 Parsimonious-cover alternatives and boundary variation

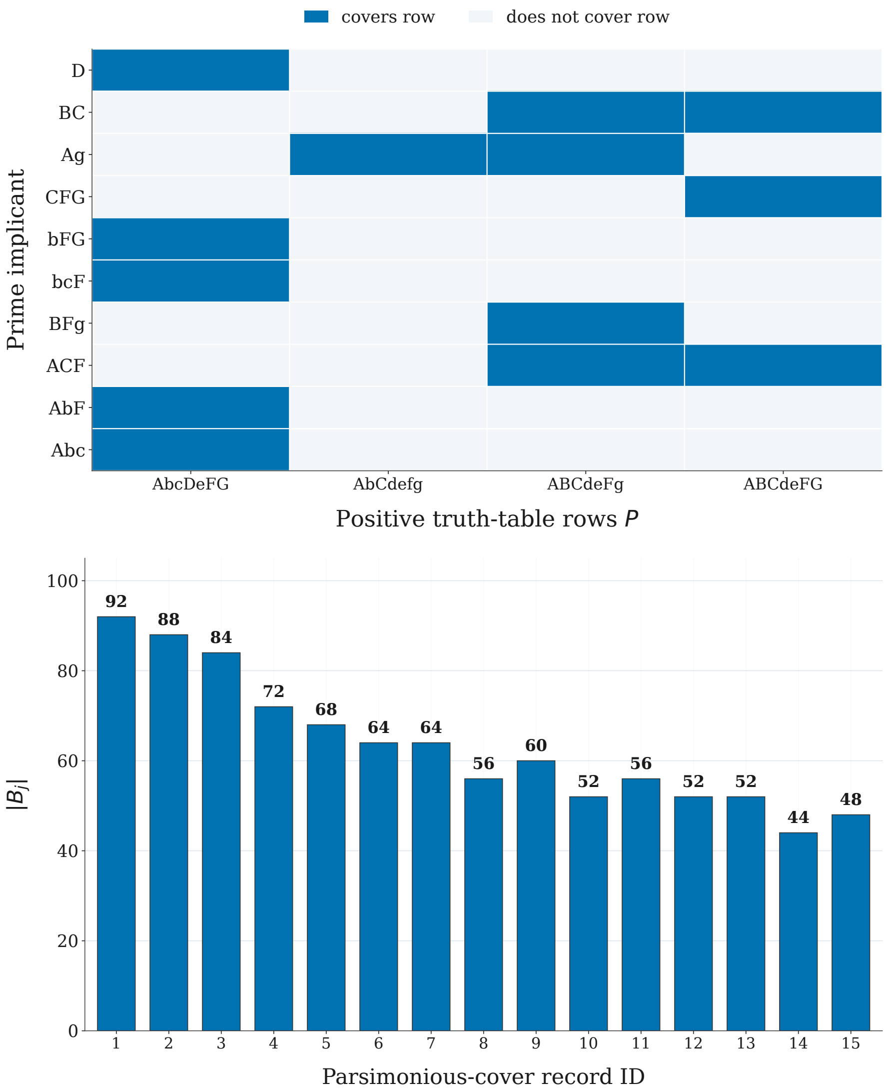

Fig. S3.1 Parsimonious-cover alternatives and boundary variation. The upper panel summarizes coverage relations between prime implicants and positive truth-table rows. The lower panel reports the selected parsimonious-boundary size for each parsimonious-cover record.

Table S3.1: Full list of parsimonious-cover alternatives

ID	Parsimonious cover S_j	$\lvert B_j \rvert$
1	D + BC + Ag	92
2	D + Ag + CFG	88
3	D + Ag + ACF	84
4	BC + Ag + bFG	72
5	BC + Ag + bcF	68
6	BC + Ag + AbF	64
7	BC + Ag + Abc	64
8	Ag + CFG + bFG	56
9	Ag + CFG + bcF	60
10	Ag + CFG + AbF	52
11	Ag + CFG + Abc	56
12	Ag + bFG + ACF	52
13	Ag + bcF + ACF	52
14	Ag + ACF + AbF	44
15	Ag + ACF + Abc	48

S3.2 Unique intermediate expressions

Table S3.2: Unique intermediate expressions associated with cover-pair records under the fixed audit inputs

Intermediate expression	Records	Cover-pair record IDs	G_{ij}^{A} sizes	G_{ij}^{B} sizes
ACg + ABCF + AbDFG	4	4, 6, 12, 14	0	2
ACg + ABCF + AbcDFG	4	5, 7, 13, 15	0, 2	2, 4
ACg + ADFG + ABCF	2	1, 3	0	0
ACg + ADFG + ABCeF	1	2	1	0
ACg + AbDFG + ABCdeF	2	8, 10	3	2
ACg + AbcDFG + ABCdeF	2	9, 11	5	2

S3.3 Path stability

Table S3.3: Path stability across cover-pair records

Path	Pattern	Records	Class	Cover-pair record IDs
ACg	1-1—0	15/15	stable core	1–15
ADFG	1–1-11	3/15	cover-sensitive	1, 2, 3
ABCF	111–1-	10/15	cover-sensitive	1, 3, 4, 5, 6, 7, 12, 13, 14, 15
AbDFG	10-1-11	6/15	cover-sensitive	4, 6, 8, 10, 12, 14
ABCeF	111-01-	1/15	cover-sensitive	2
AbcDFG	1001-11	6/15	cover-sensitive	5, 7, 9, 11, 13, 15
ABCdeF	111001-	4/15	cover-sensitive	8, 9, 10, 11

S3.4 Minterm-status stability

Table S3.4: Minterm-stability summary

Row type	Minterms	Stable support membership	Stable outside support	Sensitive support membership
P	4	4	0	0
N	8	0	8	0
R	116	15	94	7
All	128	19	102	7

Here, support denotes $P \cup E_{ij}$ for a cover-pair record.

Table S3.5: Sensitive minterms

Minterm	Type	Admitted	Not admitted	Records admitted
AbCDeFG	R	9	6	1, 2, 3, 4, 6, 8, 10, 12, 14
AbCDEFG	R	9	6	1, 2, 3, 4, 6, 8, 10, 12, 14
ABcDeFG	R	3	12	1, 2, 3
ABcDEFG	R	3	12	1, 2, 3
ABCdEFG	R	10	5	1, 3, 4, 5, 6, 7, 12, 13, 14, 15
ABCDeFG	R	11	4	1, 2, 3, 4, 5, 6, 7, 12, 13, 14, 15
ABCDEFG	R	11	4	1, 2, 3, 4, 5, 6, 7, 12, 13, 14, 15

S3.5 Admitted, G^A, and G^B status table

The remaining 94 logical remainders are consistently not directionally reachable across all 15 cover-pair records and therefore never enter the admitted-remainder sets E_{ij}. The complete cover-pair status matrix is provided in the audit-output workbook listed in Supplementary Material S5. Table S3.6 summarizes the audit-relevant remainder status counts.

Table S3.6: Audit-relevant remainder status counts across cover-pair records

Remainder	Stability class	Admitted	G^A	G^B	Not reachable
AbcDEFG, AbCdeFg, AbCdEfg, AbCdEFg, AbCDefg, AbCDeFg, AbCDEfg, AbCDEFg, ABCdefg, ABCdEfg, ABCdEFg, ABCDefg, ABCDeFg, ABCDEfg, ABCDEFg	stably admitted	15	0	0	0
AbCDeFG, AbCDEFG	variable	9	4	2	0
ABcDeFG, ABcDEFG	variable	3	0	12	0
ABCdEFG	variable	10	5	0	0
ABCDeFG, ABCDEFG	variable	11	4	0	0

S4 Additional Public Case Check: Classical Lipset Fuzzy-Set Case

S4.1 Case source and scope

Supplementary Material S4 applies the transparency report to the classical Lipset fuzzy-set example, with survival of democracy as the outcome and economic development, urbanization, literacy, industrialization, and government stability as conditions. This public-data check shows how the same reporting objects behave in a widely used QCA example beyond the school digital transformation case.

The Lipset check starts from a fixed truth-table row-level input and reports the resulting transparency-report objects under the supporting implementation. Calibration, threshold selection, truth-table construction, and the standard substantive interpretation of the Lipset example remain outside this supplementary check.

S4.2 Fixed audit inputs

The fixed truth-table input is provided as `case2_true_table.csv`. The input uses the conditions DEV, URB, LIT, IND, and STB, the outcome SURV, and a frequency column named `number`. Under the fixed row-level input used by the report, $|\Omega| = 32$, $|P| = 2$, $|N| = 7$, and $|R| = 23$. The directional-expectation vector is 11111.

The supporting implementation records one complex-cover alternative, one parsimonious-cover alternative, and one cover-pair reporting record. The complex expression is $\mathrm{DEV}* \sim \mathrm{URB} * \mathrm{LIT}* \sim \mathrm{IND} * \mathrm{STB} + \mathrm{DEV} * \mathrm{URB} * \mathrm{LIT} * \mathrm{IND} * \mathrm{STB}$. The parsimonious expression is $\mathrm{URB} * \mathrm{STB} + \mathrm{DEV}* \sim \mathrm{IND}$, with $|B_1| = 14$. These inputs define the reporting objects summarized below.

S4.3 Transparency report summary

Table S4.1 summarizes the core report objects for the public-case check. The direction-conflict label is Warning because the positive row $\mathrm{DEV}* \sim \mathrm{URB} * \mathrm{LIT}* \sim \mathrm{IND} * \mathrm{STB}$ is directionally linked to the observed negative row $\mathrm{DEV}* \sim \mathrm{URB} * \mathrm{LIT} * \mathrm{IND} * \mathrm{STB}$ under the all-positive direction vector. The expression/path-stability label is Pass because $K = 1$ and $L = 1$. The unique intermediate expression is $\mathrm{DEV} * \mathrm{LIT}* \sim \mathrm{IND} * \mathrm{STB} + \mathrm{DEV} * \mathrm{URB} * \mathrm{LIT} * \mathrm{STB}$, and both paths appear as stable core paths in the single cover-pair record.

Table S4.1 Transparency report summary for the Classical Lipset public-case check

Object	Result
Direction-conflict label	Warning; $\mathrm{DEV}* \sim \mathrm{URB} * \mathrm{LIT}* \sim \mathrm{IND} * \mathrm{STB}$ reaches $\mathrm{DEV}* \sim \mathrm{URB} * \mathrm{LIT} * \mathrm{IND} * \mathrm{STB}$ under $d = 11111$.
Cover-pair records	$K = 1$ cover-pair reporting record.
Unique intermediate expressions	$L = 1$: $\mathrm{DEV} * \mathrm{LIT}* \sim \mathrm{IND} * \mathrm{STB} + \mathrm{DEV} * \mathrm{URB} * \mathrm{LIT} * \mathrm{STB}$.
Stable core paths	$\mathrm{DEV} * \mathrm{LIT}* \sim \mathrm{IND} * \mathrm{STB}$ and $\mathrm{DEV} * \mathrm{URB} * \mathrm{LIT} * \mathrm{STB}$, each appearing in 1/1 record.
Remainder-status objects	One admitted remainder, $\mathrm{DEV} * \mathrm{URB} * \mathrm{LIT}* \sim \mathrm{IND} * \mathrm{STB}$; $\lvert G^A_{11}\rvert = \lvert G^B_{11}\rvert = 0$.
Minterm-status stability	No sensitive admitted-remainder status; $P = 2$, $N = 7$, $R = 23$, and all 32 rows have stable status.

The Lipset check therefore shows a compact transparency-report pattern: a direction-conflict warning is present, while cover-pair expression variation and minterm-status sensitivity are absent.

S4.4 Interpretation of the public-case check

The Lipset check demonstrates portability and reproducibility in a widely used public QCA example. Its value lies in showing a reporting pattern that differs from the school digital transformation case: the central issue is directional conflict rather than cover-pair or minterm-status sensitivity.

This public-data demonstration remains limited to one fixed truth-table input. Broader validation requires additional public datasets, multi-software comparisons, boundary-case tests, and versioned releases of the supporting implementation.

S5 Supplementary Data Files

The supplementary data files provide the fixed inputs and audit outputs for the two demonstrations. The file `case1_true_table.csv` provides the fixed truth-table input used in the school digital transformation case demonstration. The file `case2_true_table.csv` provides the fixed truth-table input used in the Classical Lipset fuzzy-set public-case check. The files `case1_fsqca_logical_remainder_audit_report.xlsx` and `case2_fsqca_logical_remainder_audit_report.xlsx` provide the complete audit outputs generated by the supporting implementation. Figures and tables in the main article and in Supplementary Materials S3–S4 summarize selected report objects from these files.

Table S5.1 Supplementary data files

File	Role	Contents
`case1_true_table.csv`	Fixed truth-table input for Case 1	The prepared truth-table input used in the school digital transformation case demonstration.
`case1_fsqca_logical_remainder_audit_report.xlsx`	Complete audit output for Case 1	The complete output generated by the supporting implementation for the school digital transformation case demonstration, including cover-pair results, unique intermediate expressions, path-stability records, minterm-stability records, and remainder-status records.
`case2_true_table.csv`	Fixed truth-table input for the public-case check	The prepared truth-table input used in the Classical Lipset fuzzy-set public-case check.
`case2_fsqca_logical_remainder_audit_report.xlsx`	Complete audit output for the public-case check	The complete output generated by the supporting implementation for the Classical Lipset fuzzy-set public-case check, including cover-pair results, unique intermediate expressions, path-stability records, minterm-stability records, and remainder-status records.

Together, these files provide the fixed inputs and audit outputs needed to inspect and reproduce the case-level logical-remainder transparency reports reported in the main article and Supplementary Materials.